Abstract (English)

In this thesis we wish to prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian field on the Vicsek set fractal for a space of suitable test functions.

We will introduce the notion of self-similar set and their properties so as to construct the Vicsek. We will also examine the concepts of Hausdorff, walk and spectral dimension will be introduced, followed by a normalized Hausdorff measure for the Vicsek set.

Subsequently we construct the discrete Laplacian, the Kigami Laplacian and the Dirichlet Laplacian, followed by the introduction of the discrete Riesz kernel and the fractional Riesz kernel this will allow us to define the discrete fractional Laplacian.

Further notions of functional analysis are needed to construct and define semi-groups and heat kernels for densely defined self-adjoint operators. These will in turn be used in the definition of the fractional Laplacian. A space of suitable test functions will also be defined for the fractional Laplacian.

Finally both discrete fractional Gaussian fields and fractional Gaussian fields are introduced along with their properties leading us to prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields on a space of suitable test functions.

Resumé (Dansk)

I dette speciale ønsker vi at vise konvergens i fordeling af diskrete fraktionerede Gaussiske felter til fraktionerede Gaussiske felter på Vicsek fraktalet for en passende mængde af testfunktioner.

Først vil vi introducere idéen omkring selv-similære mængder, deres egenskaber og konstruere Vicsek fractalet ud fra disse idéer. Dernæst vil koncepterne Hausdorff, spectral og walk-dimension blive introduceret, efterfulgt af et normalizeret Hausdorff mål på Vicsek fractalet.

Herefter konstruerer vi den diskrete Laplacian, Kigami Laplacianen og Dirichlet Laplacianen, som vil blive fulgt af en introduktion til den diskrete fraktionerede Riesz kerne og den fraktionerede Riesz kerne, hvilket lader os definere den diskrete fraktionerede Laplacian.

Yderligere begreber fra functional analyse vil blive brugt til at konstruere semi-grupper og en heat-kernel for tæt definerede selvadjungerede operatorer, hvilket er nødvendigt for konstruktionen af den fraktionelle Laplacian. Samtidigt, vil en passende mængde af testfunktioner blive defineret til den fraktionelle Laplacian.

Sidst vil både diskrete fraktionerede Gaussiske felter og fraktionerede Gaussiske felter blive introduceret sammen med deres egenskaber. Dette leder os til at bevise konvergens i fordeling for diskrete fraktionerede Gaussiske felter til fraktionerede Gaussiske felter på Vicsek fraktalet for en passende mængde af testfunktioner.

Contents

Chapter 1

Introduction

The notion of fractals was introduced in the 1970's by Benoît B. Mandelbrot, where he defined it as an object that has Hausdorff dimension strictly larger than topological dimension. Many have since built upon this definition of fractals and greatly expanded the area of fractal analysis [12], [14], [23], [20]. The thesis will expand upon this by introducing and studying a family of Gaussian fields, their discrete analogues and the convergence in distribution of the discrete field on the Vicsek set fractal.

The thesis is an extension of the paper [4] by Fabrice Baudoin and Li Chen. It will follow the aforementioned paper, but will diverge slightly, by studying the Vicsek set rather than the Sierpinski gasket.

The Vicsek set is defined as the non-empty compact set K such that

$$K = \bigcup_{i=1}^{5} \mathcal{S}_i(K), \quad \text{where} \quad \mathcal{S}_i(x) = \frac{1}{3}(x - p_i) + p_i, \quad i = 1, 2, \ldots 5.$$

Here $p_i \in V_0$ are the edge points of a "plus" with a point in the middle (see Figure 1.1). The discrete Vicsek is then defined recursively as

$$V_m = \bigcup_{i=1}^{5} \mathcal{S}_i(V_{m-1}).$$

We can then create a discrete graph of the Vicsek set as shown below.

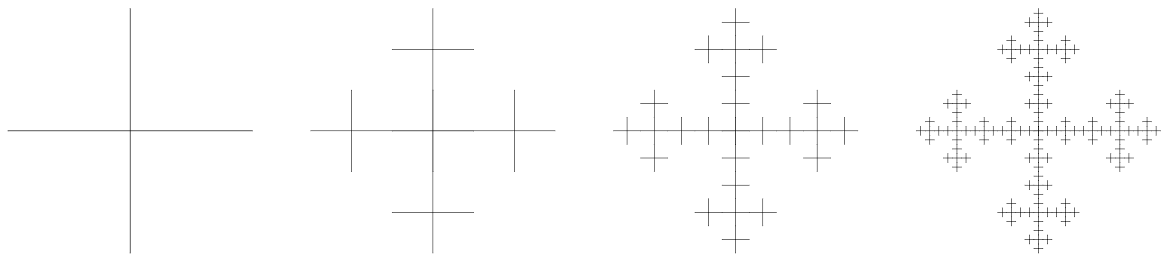

Figure 1.1: Vicsek graph, G_0, G_1, G_2 and G_3.

As stated, the goal of this thesis is to study the fractional Gaussian fields. So if we let $K \subset \mathbb{R}^2$ be the Vicsek set fractal, then the family of Gaussian fields we wish to study are defined in K

indexed by a parameter $s \geq 0$ satisfying

$$\mathbb{E}\left[X_s(f)X_s(g)\right] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \ \mathrm{d}\mu$$

where f, g are in a space of suitable test functions on K, μ is a normalized Hausdorff measure and Δ is the Dirichlet Laplacian on K. Such fields are heuristically defined as the distribution $X_s = (-\Delta)^{-s} W$ where W is a white noise on $L^2(K, \mu)$.

Likewise the discrete fractional Gaussian fields are heuristically given by $X_s^m = (-\Delta_m)^{-s} W_i$ where Δ_m is the discrete Laplacian and W_i is a sequence of i.i.d Gaussian random variables with mean zero and variance one. These are precisely the fields of which we will prove convergence in distribution.

As we will be studying the Vicsek set, it is natural that we start by constructing the set and proving some of its basic properties. While we have described the Vicsek set as a non-empty compact self-similar set, we will first define and prove some basic properties of such self-similar sets.

1.1 Iterated Function System

The iterated functions systems (IFS) gives us a way to work with self-similar sets. Constructing the IFS also allows us to prove that the Vicsek set actually exists. Thus we start by defining the iterated function systems.

Definition 1.1: A finite family of functions $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$ with $m \geq 2$ is called an *iterated function system* (IFS) if there exists $\{c_i\}_{1 \leq i \leq m} \in (0,1)$ such that all $\mathcal{S}_i$ are *contractions*. This is exactly where

$$\left|\mathcal{S}_i(x) - \mathcal{S}_i(y)\right| \leq c_i |x - y| \, .$$

Furthermore we call a non-empty compact subset K of X an *invariant set* for the IFS if

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.1}$$

This then begs the question, if there for every IFS exists an invariant set? We shall see that this is indeed true, but to be able to prove that a unique invariant set exists, we will need to define the Hausdorff metric.

Definition 1.2: Define $A_\delta = \{x \in \mathcal{A} : \exists a \in A, |x - a| < \delta\}$ The *Hausdorff metric* is given by

$$d_H(X, Y) = \inf \left\{\delta : X \subset Y_\delta \text{ and } Y \subset X_\delta\right\}$$

The proof that d_H is a metric can be found in [18].

Theorem 1.3: Consider the IFS given by $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$, then there is a unique invariant set K for the IFS. Moreover, if we define a transformation S, on the class $\mathcal{A}$ of non-empty compact sets, by

$$S(E) = \bigcup_{i=1}^{m} \mathcal{S}_i(E), \quad \text{for } E \in \mathcal{A}$$

and write S^k to be the k'th iterate of S, so $S^0 = E$ and $S^k(E) = S(S^{k-1}(E))$ for $k \geq 1$, then

$$F = \bigcap_{i=0}^{\infty} S^k(E)$$

for every set $E \in \mathcal{A}$ such that $S_i(E) \subset E$ for all i.

Proof. Let E be any set in $\mathcal{A}$ such that $S_i(E) \subset E$ for all i. By induction we find that since $S^0 = E \supset S(E) = S^1(E)$, then $S^{k+1}(E) = S(S^k(E)) \subset S(S^{k-1}(E)) = S^k(E)$, by the induction assumption $S^k \subset S^{k-1}$, in which case we get $S^k(E) \subset S^{k-1}(E)$. This in turn implies that $S^k(E)$ is a decreasing sequence of non-empty compact sets, in which case it must have a non-empty compact intersection $K = \bigcap_{k\geq 0} S^k(E)$. Since $S^k(E)$ is a decreasing sequence, then $S(K) = K$ satisfying (1.1).

Now moving on to uniqueness, we will need the following inequality

$$d_H(S(A), S(B)) = d_H\left(\bigcup_{i=1}^{m} \mathcal{S}_i(A), \bigcup_{i=1}^{m} \mathcal{S}_i(B)\right) \leq \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A, B). \tag{1.2}$$

Assuming A and B are both invariant sets of the same IFS, then $S(A) = A$ and $S(B) = B$, giving us that

$$d_H(A, B) = d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A, B).$$

Since $0 < \max_{1\leq i\leq m} c_i < 1$, then $d_H(A, B) = 0$ whereby $A = B$ making the invariant set unique.

Finally we will prove (1.2). The first inequality comes by choosing

$$\delta = \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B))$$

since it follows that $\mathcal{S}_i(B) \subset (\mathcal{S}_i(A))_\delta$ for all $0 \leq i \leq m$. Thus it will definitely hold for $\bigcup_{i=1}^m \mathcal{S}_i(B) \subset \left(\bigcup_{i=1}^m \mathcal{S}_i(A)\right)_\delta$ and vice versa.

The second inequality comes from letting $\delta = d(A, B)$. Then for every $a \in A$ we can choose some $b \in B$ such that $|a - b| \leq \delta$ so that $\left|\mathcal{S}_i(a) - \mathcal{S}_i(b)\right| \leq c_i\delta$ and whereby $\mathcal{S}_i(A) \subset \left(\mathcal{S}_i(B)\right)_{c_i\delta}$. □

A consequence of Theorem 1.3. The theorem gives us a way to create a (not necessarily unique) sequence $(i_1, i_2, \ldots)$, where $1 \leq i_j \leq m$, such that for each $x \in F$ we can write

$$\{x\} = \{x_{i_1, i_2, \ldots}\} = \bigcap_{k\geq 0} K_{i_1, \ldots i_k} \tag{1.3}$$

where $K_{i_1,\ldots i_k} = (\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_k}(K)$. Thus in a way we give each $x \in F$ an address. This is possible since S^k is decreasing as k increases.

If we further let $\mathcal{I}$ denote all sequences of the type $(i_1, i_2, \ldots)$, then we can describe K as

$$K = \bigcup_{\mathcal{I}} \{x_{i_1, \ldots, i_k}\}. \tag{1.4}$$

1.2 Construction of the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in $\mathbb{R}^2$ given by $p_1 = (0,0), p_2 = (\xi, 0)$, $p_3 = (0, -\xi), p_4 = (-\xi, 0)$ and $p_5 = (0, \xi)$ for some $\xi > 0$. Then define

$$\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \ldots 5.$$

Where $\{\mathcal{S}_1, \ldots, \mathcal{S}_5\}$ indeed is an IFS, since for every $x, y \in \mathbb{R}^2$ we have

$$|\mathcal{S}_i(x) - \mathcal{S}_i(x)| = \left| \frac{1}{3}(x - p_i) + p_i - \left(\frac{1}{3}(y - p_i) + p_i \right) \right| = \frac{1}{3}|x - y|\,.$$

Thus by Theorem 1.3 we define the Vicsek set K as the unique non-empty compact subset in $\mathbb{R}^2$ such that

$$K = \bigcup_{i=1}^{5} \mathcal{S}_i(K).$$

Since we are going to study the convergence of discretely defined operators we can define a discrete Vicsek set in the following way. We start by defining a sequence of sets $(V_m)_{m\geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^{5} \mathcal{S}_i(V_m)$$

This gives us a sequence of Vicsek set graphs $(G_m)_{m\geq 0}$ whose vertices are $(V_m)_{m\geq 0}$.

Notation: We denote $V_* = \cup_{m\geq 0} V_m$ and for any $p \in V_m$ we write the collection of neighbours of p in G_m by $V_{m,p}$. We may write $x \underset{m}{\sim} y$ when x and y are neighbours and $x \sim y$, when there is no confusion as to what V_m, x and y belongs to. Lastly, note that $N_m = \#V_m = 5^m \cdot 4 + 1$.

1.3 Hausdorff Dimension

The Hausdorff dimension can be thought of as a form of "fractal dimension" and will be important as it will be used to bound and define many of the mathematical objects that we will manipulate.

The Hausdorff dimension has the advantage of being defined for any set and it is based on measures, which we know how to work with. The greatest disadvantage is that it is very difficult to calculate from the definition, in most cases. It will be quite easy to calculate, if we are given an invariant set of an IFS. We begin this section by defining a cover for sets.

Definition 1.4: Let X be a metric space with metric $d(\cdot,\cdot)$. We say the countable collection of set $\{U_i\}_{i\geq 1}$ is a δ-*cover* of K if $K \subset \bigcup_{i\geq 1} U_i$ and $\operatorname{diam} U_i \leq \delta$, where $\operatorname{diam}(A) = \sup_{x,y\in A} d(x,y)$.

Definition 1.5: Let X be a metric space. If $K \subset X$ and $d \in [0, \infty)$ we define

$$H^d_\delta(K) = \inf \left\{ \sum_{i=1}^{\infty} \operatorname{diam}(U_i)^d \ : \ \bigcup_{i=1}^{\infty} U_i \supseteq K, \operatorname{diam}(U_i) < \delta \right\}.$$

Then the d-dimensional *Hausdorff measure* is given by

$$\mathcal{H}^d(K) = \lim_{\delta\to 0} H^d_\delta(K)$$

Remark: The Hausdorff measure defined above, is not a measure in the normal measure theory sense. Rather it is an outer measure, although it is possible to prove that the Hausdorff measure restricted to Borel-sets, is a measure.

Moving swiftly along, we now want to investigate the behaviour of $\mathcal{H}^s$ as s changes. Let $\{U_i\}_{i\geq 1}$ be a δ-cover of K, then we have for $0 \leq s < t$ that

$$\sum_{i\geq 1} \operatorname{diam}(U_i)^t = \sum_{i\geq 1} \operatorname{diam}(U_i)^{t-s} \operatorname{diam}(U_i)^s \leq \delta^{t-s} \sum_{i\geq 1} \operatorname{diam}(U_i)^s.$$

Taking infimum on both sides we get that $H^t_\delta(K) \leq \delta^{t-s} H^s_\delta(K)$. If we let $\delta \to 0$ we see that if $\mathcal{H}^s(K) < \infty$ then $\mathcal{H}^s(K) = 0$ since $\delta^{t-s} \to 0$. Likewise if $\mathcal{H}^t > 0$ then $\mathcal{H}^s = \infty$. This tells us that if there is a point, say κ, such that $\mathcal{H}^\kappa$ is both positive and finite, then $\mathcal{H}^s$ will be 0 for all $s > \kappa$ and ∞ for all $s < \kappa$. Such a value is called the *Hausdorff Dimension* of K.

Definition 1.6: The *Hausdorff dimension*, denoted $d_h(K)$ is given by

$$d_h(K) = \inf\left\{s \geq 0 : \mathcal{H}^s(K) = 0\right\} = \sup\left\{s \geq 0 : \mathcal{H}^s(K) = \infty\right\}.$$

Notation: When there is no confusion with regards to what set we are working with, we denote the Hausdorff dimension by d_h.

As stated before, it is quite difficult to determine the Hausdorff dimension from the definition. Fortunately, we can with a few assumptions find the Hausdorff dimension quite easily. But before we can prove this theorem, we first define the open set condition and state a few important lemmas.

Definition 1.7: We say that an IFS $\{\mathcal{S}_i\}_{1\leq i\leq m}$, satisfies the *open set condition* if there exists a non-empty bounded open set V such that

$$V \supset \bigcup_{i=1}^{m} \mathcal{S}_i(V)$$

and each $\mathcal{S}_i(V)$ is disjoint.

The following Lemma 1.8 describes a lower bound for the Hausdorff dimension.

Lemma 1.8: Let μ be a measure such that $\mu(F) \in (0, \infty)$ and suppose there are numbers $c, \varepsilon > 0$ such that for some $s \geq 0$

$$\mu(U) \leq c \operatorname{diam}(U)^s$$

for all sets U with $\operatorname{diam}(U) \leq \varepsilon$. Then $\mathcal{H}^s(F) \geq \frac{\mu(F)}{c}$ and $s \leq d_h$

Proof. If $\{U_i\}_{i\in I}$ is any cover of F then

$$0 < \mu(F) \leq \mu\left(\bigcup_{i\in I} U_i\right) \leq \sum_{i\in I} \mu(U_i) \leq c \sum_{i\in I} \operatorname{diam}(U_i)^s$$

by definition of measures and properties of μ. Taking infimum we get $\mathcal{H}^s_\delta \geq \frac{\mu(F)}{c}$ and therefore $\mathcal{H}^s \geq \frac{\mu(F)}{c} > 0$ so by Definition 1.6 $s \leq d_h$. $\square$

Lemma 1.9: Let $\{V_i\}$ be a collection of disjoint open subsets of $\mathbb{R}^n$ such that each V_i contains a ball of radius $a_1 r$ and is contained in a ball of radius $a_2 r$. Then any ball B of radius R intersects at most $(1+2a_2)^n a_1^n$ of the closures $\overline{V}_i$

Proof. By assumption we know that $\operatorname{diam}(V_i) \le 2a_2 r$ for all V_i. So if some $\overline{V}_i$ intersects the ball with radius r, then $\overline{V}_i$ must be contained in the ball with radius $(2a_2+1)r$ concentric with the ball with radius r. Now let q denote the number of closures $\overline{V}_i$ such that $\overline{V}_i \cap B(x,r) \neq \emptyset$. Then there exists q disjoint balls with radius $a_1 r$ that are contained in the ball with radius $(2a_2+1)r$, since by assumption, each $\overline{V}_i$ contains a ball of radius $a_1 r$. It follows that $q(a_1 r)^n \le (2a_2+1)^n r^n$ which gives us the bound for q. $\square$

Combing the above results, we are able to state the following theorem:

Theorem 1.10: Suppose Definition 1.7 holds for the similarities $\mathcal{S}_i$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$ for $1 \le i \le m$. If, for K the following holds

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.5}$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^{m} c_i^s = 1. \tag{1.6}$$

Moreover, for this value of s, $0 < \mathcal{H}^s(K) < \infty$.

Proof. Let $\mathcal{I}_k$ denote the set of all k-term sequences $(i_1, \ldots, i_k)$ with $1 \le i_j \le m$. For any set A and any sequence $(i_1, \ldots, i_k) \in \mathcal{I}_k$ we denote $A_{i_1,\ldots,i_k} = (\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_k})(A)$.

We will start by finding an upper bound for the Hausdorff measure. We do this by finding an appropriate δ-cover for K. It follows that by repeated use of (1.5) that

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) = \bigcup_{i=1}^{m} \mathcal{S}_i \left(\bigcup_{j=1}^{m} \mathcal{S}_j(K) \right) = \bigcup_{i,j=1}^{m} (\mathcal{S}_i \circ \mathcal{S}_j)(K) = \ldots = \bigcup_{\mathcal{I}_k} \mathcal{S}_{i_1,\ldots,i_k}(K) = \bigcup_{\mathcal{I}_k} K_{i_1,\ldots,i_k}.$$

Now noting that $\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_k}$ has the ratios $c_{i_1} \cdots c_{i_k}$, we see

$$\sum_{\mathcal{I}_k} \operatorname{diam}(K_{i_1,\ldots i_k})^s = \sum_{\mathcal{I}_k} (c_{i_1} \ldots c_{i_k})^s \operatorname{diam}(K)^s = \left(\sum_{i_1} c_{i_1}^s \right) \cdots \left(\sum_{i_k} c_{i_k}^s \right) \operatorname{diam}(K)^s = \operatorname{diam}(K)^s$$

where the last equality comes from (1.6). Now for any $\delta > 0$ we can choose k so

$$\operatorname{diam}(K_{i_1,\ldots i_k}) \le \left(\max_{1 \le i \le m} c_i \right)^k \operatorname{diam}(K) \le \delta$$

since $\max_{1 \le i \le m} c_i \in (0,1)$. Thus $\{K_{i_1,\ldots i_k}\}_{\mathcal{I}_k}$ is a δ-cover for K whereby $\mathcal{H}^s_\delta(K) \le \operatorname{diam}(K)^s$ in which case $\mathcal{H}^s(K) \le \operatorname{diam}(K)^s$ giving an upper bound.

Next we need a lower bound for the Hausdorff dimension, which is substantially more difficult. Let $\mathcal{I} = \{(i_1, i_2, \ldots) : 1 \le i_j \le m\}$ be the set of all infinite sequences and let

$$\mathcal{I}_{i_1,\ldots i_k} = \{(i_1, \ldots, i_k, q_{k+1}, \ldots) : 1 \le q_j \le m\},$$

so the first initial k "terms" are chosen and then the rest are the remaining permutations.

Now we create an outer measure μ on subset of $\mathcal{I}$ defined by $\mu(\mathcal{I}_{i_1,\dots i_k}) = (c_{i_1}\dots c_{i_k})^s$. From this outer measure μ we construct an outer measure on K named $\tilde{\mu}$ defined in the following way. For some subset $A \subseteq K$

$$\tilde{\mu}(A) = \mu\left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1,i_2,\dots} \in A \cap K\}\right)$$

The proof that μ and $\tilde{\mu}$ are indeed outer measures can be found in [17, Section 7].

Now we want to show that $\tilde{\mu}$ fulfils the properties of Lemma 1.8.
Let V be the open set that satisfies Definition 1.7. Since $\overline{V} \supset \mathcal{S}(\overline{V}) = \cup_{i=1}^m \mathcal{S}_i(\overline{V})$, then by Theorem 1.3 the decreasing sequence $S^k(\overline{V})$ converges to K. In particular we have $\overline{V} \supset S(\overline{V}) \supseteq \cup_{i=1}^m \mathcal{S}_i(\overline{V}) = F$ and $\overline{V}_{i_1,\dots i_k} = (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(\overline{V}) \supseteq (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(\overline{F}) = F_{i_1,\dots i_k}$ for each finite sequence $(i_1, \dots, i_k)$.

Now let B be any ball $r < 1$. We estimate $\tilde{\mu}(B)$ by considering sets $V_{i_1,\dots i_k}$ with diameters comparable with that of B with closures intersecting $F \cap B$. Further we define $\mathcal{Q}$ as the set of sequences $(i_1, \dots, i_k) \in \mathcal{I}$ such that i_k is the smallest k for which the following holds

$$\left(\min_{1\le i\le m} c_i\right) r \le c_{i_1}\dots c_{i_k} \le r.$$

Note that no sequence in $\mathcal{Q}$ extends to another sequence. Since, if two sequences are the same, but one is longer than the other, the shorter sequence would be the one in $\mathcal{Q}$. Now by Definition 1.7 $V_1, \dots, V_m$ are disjoint and so are $V_{i_1,\dots i_k,1}, \dots, V_{i_1,\dots i_k,m}$ for each $(i_1, \dots, i_k)$. In particular we have that $\{V_{i_1,\dots i_k} : (i_1, \dots, i_k) \in \mathcal{Q}\}$ is a collection of disjoint sets. We further note that by (1.4) we have

$$K = \bigcup_{\mathcal{I}} \{x_{i_1,i_2,\dots}\} \subseteq \bigcup_{\mathcal{Q}} K_{i_1,\dots,i_k} \subseteq \bigcup_{\mathcal{Q}} \overline{V}_{i_1,\dots,i_k}. \tag{1.7}$$

Next we choose a_1 and a_2 so that V contains a ball of radius a_1 and is contained in a ball of radius a_2. Then for all $(i_1, \dots i_k) \in \mathcal{Q}$ the set $V_{i_1,\dots i_k}$ contains a ball of radius $c_{i_1}\dots c_{i_k}a_1$ and therefore one of radius $(\min_{1\le i\le m} c_i)\, a_1 r$. Furthermore it is contained in a ball of radius $c_{i_1}\dots c_{i_k}a_2$ and therefore one of radius $a_2 r$.

Now let $\mathcal{Q}_B$ be the set of all sequences $(i_1, \dots, i_k) \in \mathcal{Q}$ such that $\overline{V}_{i_1,\dots i_k}$ intersects B. By Lemma 1.9 there are at most

$$q = (1 + 2b)^n \left(\min_{1\le i\le m} c_i\right)^{-n} a^{-n}$$

sequences in $\mathcal{Q}_B$. Thus we find that if

$$x_{i_1,\dots i_k} \in F \cap B \subseteq \bigcup_{\mathcal{Q}_B} \overline{V}_{i_1,\dots i_k}, \qquad \text{by (1.7)},$$

then there must be some k such that $(i_1, \dots, i_k) \in \mathcal{Q}_B$ and therefore $(i_1, i_2, \dots) \in \cup_{\mathcal{Q}_B} I_{i_1,\dots i_k}$. We can therefore conclude

$$\tilde{\mu}(B) = \mu\left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1,i_2,\dots} \in F \cap B\}\right) \le \mu\left(\bigcup_{\mathcal{Q}_B} I_{i_1,\dots,i_k}\right),$$

whereby it follows that

$$\tilde{\mu}(B) \le \sum_{\mathcal{Q}_B} \mu\left(I_{i_1,\dots,i_k}\right) = \sum_{\mathcal{Q}_B} \left(c_{i_1}\dots c_{i_k}\right)^s \le \sum_{\mathcal{Q}_B} r^s \le q r^s.$$

Since any set U is contained in a ball of radius $\operatorname{diam}(U)$, we have that $\tilde{\mu} \le q \operatorname{diam}(U)^s$. By Lemma 1.8 we have that $\mathcal{H}^s(F) \ge \frac{\tilde{\mu}(F)}{q} = q^{-1} > 0$, thus $0 < q^{-1} \le \mathcal{H}^s(F) \le \operatorname{diam}(F)^s < \infty$ such that by Definition 1.6 we have that $s = d_h$ □

1.3.1 Spectral and Walk dimension

Further in this thesis we will bound certain random fields and functions by a constant we call the spectral dimension. It depends on two other factors, namely the Hausdorff dimension, which we have just discussed, and of the walk dimension. Properly explaining the walk dimension is outside the scope of this thesis, but we will give some intuition as to what it is.

Let B_t be a Brownian motion on K, and $B(x,r)$ be the ball centered at $x \in K$ with radius r. Then the *walk dimension* is the constant $d_w > 0$ such that

$$c_1 r^{d_w} \leq \mathbb{E}\left[\tau(x,r)\right] \leq c_2 r^{d_w}$$

where $\tau(x,r) = \inf\left\{t \geq 0 : B_t \notin B(x,r)\right\}$. The intuition is that it is the average escape rate, meaning how fast the Brownian motion escapes the ball. This explanation for the walk dimension is a rather simplified and condensed one. For further reference see [23, Chapter 3]. It does however, allow us to define the spectral dimension as follows:

Definition 1.11: Let d_h be the Hausdorff dimension and d_w the walk dimension then the *spectral dimension* is

$$d_s = \frac{2d_h}{d_w}.$$

1.4 Hausdorff, Walk and Spectral Dimension of the Vicsek set

Since the main purpose of the thesis is to study the Vicsek set, we will in this section, state the different dimensions needed.

Recalling that the Vicsek set is the unique set K for the IFS $\{\mathcal{S}_1, \ldots, \mathcal{S}_5\}$ with $\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i$ where $p_i \in V_0$, it holds that

$$K = \bigcup_{i=1}^{5} \mathcal{S}_i(K).$$

We want to use Theorem 1.10 to find the Hausdorff dimension, so we need to check that $\mathcal{S}_{i1 \leq i \leq 5}$ satisfies Definition 1.7.

If we choose the open set V to be the open box where each corner is a point in V_0 (see Figure 1.2), then by continuity of $\mathcal{S}_i$ we have that $\mathcal{S}_i(V)$ is open and disjoint. The disjointness of $\mathcal{S}_i$ is that each $\mathcal{S}_i$ "retracts" the whole set towards the point p_i and we get that the sets are disjoint. One might worry about the points where the open sets meet, but since the points in V_0, excluding the middle point, are not included in the initial open box, we have that they do not intersect.

We can therefore conclude from Theorem 1.10 that

$$\sum_{i=1}^{5} \left(\frac{1}{3}\right)^{d_h} = 1 \Rightarrow \left(\frac{1}{3}\right)^{d_h} = \frac{1}{5} \Rightarrow d_h = \frac{\log 5}{\log 3}.$$

As stated earlier we will not be calculating the walk dimension, but we will state it. The walk dimension for the Vicsek set is $d_w = d_h + 1$ and we can make the following lemma.

Lemma 1.12: The following constants hold for the Vicsek set K

$$d_h = \frac{\log 5}{\log 3}, \quad d_w = d_h + 1, \quad \text{and} \quad d_s = \frac{d_h}{2d_w} = \frac{\log 5}{2\log 15}.$$

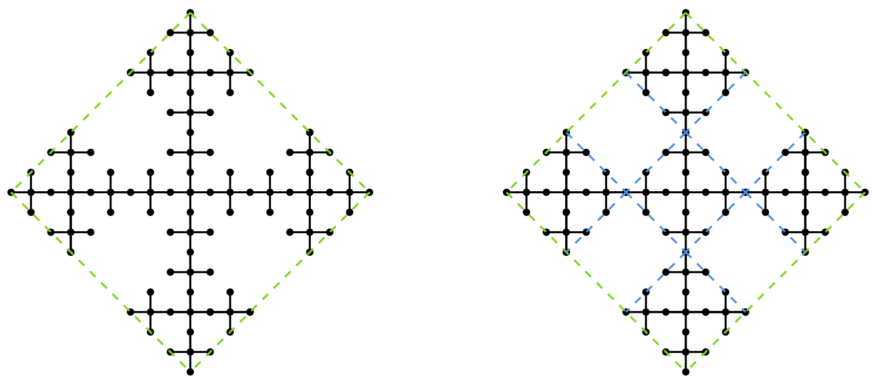

Figure 1.2: The opens sets V (green) and $\mathcal{S}_i(V)$ (blue).

Having calculated the Hausdorff dimension of the Vicsek set we will immediately use it to construct the following measure.

Definition 1.13: The *normalized Hausdorff measure* on K which is given by the Borel measure μ on K such for any $i_1, \dots i_n$, where $1 \le i_j \le 5$, we have

$$\mu\left(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)\right) = 5^{-n}.$$

The measure μ has the following property:

Proposition 1.14: The measure μ is d_h-Ahlfors regular, i.e. there exists $c_1, c_2 > 0$ such that for every $x \in K$ and $r \in [0, \operatorname{diam}(K)]$

$$c_1 r^{d_h} \le \mu(B(x,r)) \le c_2 r^{d_h}.$$

Proof. If $r = 0$ the proof is trivial. So let $r \in (0,1)$, and choose n so $R\left(\frac{2}{3}\right)^{n+1} \le r \le R\left(\frac{2}{3}\right)^{n}$ where $R = \operatorname{diam}(K)$. Note that for a sequence we have

$$\operatorname{diam}(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m}(K)) = \left(\frac{2}{3}\right)^m \cdot R.$$

We can therefore construct a sequence $i_1, \dots, i_n$ and $i_1, \dots, i_{n+1}$ such that $B(x,r) \supset \left(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{n+1}}\right)(K)$ and $\left(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}\right)(K) \cap B(x,r) \neq \emptyset$. Since $B(x,r)$ intersects with at most 5 of these sequences, we find an upper bound to be

$$\begin{aligned}\mu(B(x,r)) &\le 5 \cdot \mu\left(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)\right) = 5 \cdot 5^{-n} \\ &= 5^2 \cdot 5^{-(n+1)} \le 5 \cdot \left(R \cdot 3^{-(n+1)}\right)^{d_h} \le 5 \cdot 2^{-n} \cdot r^{d_h}\end{aligned}$$

noting that $2^{-n} r \le r$. This upper bound also works for $R < 1$ since we just omit the constant from the calculations and have $3^{-(n+1)} \le R^{-1} 2^{-(n+1)} r$ whereby the upper bound becomes $5 \cdot \left(R^{-1} 2^{-(n+1)} r\right)^{d_h}$. Likewise, for the lower bound we have that there exists a sequence $i_1, \dots i_{n+1}$ such that $B(x,r) \supset S_{i_1} \circ \dots \circ S_{i_n}$ in which case

$$\mu(B(x,r)) \ge \left(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{n+1}}(K)\right) = 5^{-(n+1)} \ge R \cdot 5^{-(n+1)} \ge 5 \cdot \left(R^{-1} 2^{-n} r\right)^{d_h}.$$

Again this lower bound also holds for $R > 1$ since we again omit R in the calculations and have $R^{-1} 2^{-n} r \le 3^{-n}$ whereby the lower bound becomes $5 \cdot \left(R^{-r} 2^{-n} r\right)^{d_h}$.

Now assume $\operatorname{diam}(K) \geq r \geq 1$. The upper bound is simply $\mu(B(x,r)) \leq \mu(K) = 1 \leq r$. For the lower bound, there exists $m \in \mathbb{N}$ so $3^{m-1} \leq r \leq 3^m$, then

$$\mu(B(x,r)) \geq \mu(\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_{m-1}}) = 5^{-(m-1)} = 5 \cdot (3^{-m})^{d_h} \geq (r3^{-2m})^{d_h}$$

as desired. □

As we will want to work with the graph approximations of the Vicsek set, we will define a sequence of discrete measures.

Definition 1.15: Let $(\mu_m)_{m\geq 0}$ be a sequence of discrete measures on V_m given by

$$\mu_m = \frac{1}{a_m} \sum_{p \in V_m} \delta_p \tag{1.8}$$

where $a_m = 5^m \cdot 4 + 1$.

We have that $\mu_m \to \mu$ as $m \to \infty$. This follows from the fact that $\mathcal{S}_i(K)$ ”shrinks” the fractal down to 1/5 the size around the point p_i, in which case for any sequence $i_1, \ldots i_n$ we have that

$$\begin{aligned}\lim_{m\to\infty} \mu_m(\mathcal{S}_{i_1}, \circ \ldots \circ \mathcal{S}_{i_n}(K)) &= \lim_{m\to\infty} \frac{1}{a_m} \sum_{p\in V_m} \delta_p \left(\left(\mathcal{S}_{i_1}, \circ \ldots \circ \mathcal{S}_{i_n}\right)(K) \right) \\ &= \lim_{m\to\infty} \frac{1}{a_m} \cdot 5^{-n} \cdot (5^m \cdot 4 + 1) = 5^{-n} = \mu(\mathcal{S}_{i_1}, \circ \ldots \circ \mathcal{S}_{i_n}(K))\end{aligned}$$

recalling that $\#V_m = 5^m \cdot 4 + 1$. This leads us to the following corollary:

Corollary 1.16: The sequence of measures $(\mu_m)_{m\geq 0}$ defined in (1.8) converges weakly to the normalized Hausdorff measure μ on the Vicsek set K i.e.

$$\lim_{m\to\infty} \int_K f \, \mathrm{d}\mu_m = \int_K f \, \mathrm{d}\mu.$$

Chapter 2

Laplacians

In Chapter 3 the Gaussian random fields will be defined using the eigenvalues and eigenfunctions of the Dirichlet Laplacian. We will find that the Dirichlet Laplacian is an extension of the Kigami Laplacian. But to construct the Kigami Laplacian we need the discrete Laplacian as we will prove that the Kigami Laplacian is the pointwise limit of the discrete Laplacian. We will therefore start by introducing the discrete Laplacian, then the Kigami Laplacian, culminating with the Dirichlet Laplacian. Therefore consider the discrete Laplacian on the Vicsek set.

Definition 2.1: Let $\ell(V_m)$ be the set of functions of $f\colon V_m \to \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = 3^m \frac{1}{N_p} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

where

$$N_p = \begin{cases} 5^{-m} & \text{if } \#V_{m,p} = 4 \\ 2 \cdot 5^{-(m+1)} & \text{if } \#V_{m,p} = 2 \\ 5^{-(m+1)} & \text{if } \#V_{m,p} = 1. \end{cases}$$

Remark 2.2: One is able to describe the discrete Laplacian of a function f as a matrix. Let $p_1, \dots p_{N_m}$ be the vertices in V_m then the *Laplacian matrix* L_m given by

$$(L_m)_{i,j} = \begin{cases} \#V_{m,p_i}, & \text{if } i = j \\ -1, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise} \end{cases}$$

which satisfies:

$$L_m F = \begin{pmatrix} \Delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

In addition $\#V_{m,p_i}$ is also called the *degree of* p_i

2.1 Graph Energies

In this section we will be following the arguments given by Strichartz in [27, Chapter 1]. Many of the proofs are going to rely on the graph energy. Thus, we will first define and prove some properties of the graph energies form.

Definition 2.3: Given a finite connected graph G_m and it vertices V_m then for functions $u, v\colon V_m \to \mathbb{R}$ we define the *graph energy* to be

$$E_m(u,v) = \sum_{x\sim y} \left(u(x) - u(y)\right)\left(v(x) - v(y)\right).$$

Notation: We will write $E_m(u) = E_m(u,u)$.

Since we want to work on the whole Vicsek set K, we will want to create an extension of a function $u\colon V_m \to \mathbb{R}$, such that it is defined on V_{m+n}. This is done in the following way:

Definition 2.4: Let $u\colon V_m \to \mathbb{R}$. Then an *extension of* u is a function $u'\colon V_{m+n} \to \mathbb{R}$ where $u' \mid_{V_m} = u$.

One quickly notices that there must be a function that minimizes the graph energy of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 2.5: Let $\tilde{u}\colon V_{m+n} \to \mathbb{R}$ with $\tilde{u} \mid_{V_m} = u$ being a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u'\colon V_{m+n} \to \mathbb{R}$ with $u' \mid_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = r^n E_m(u)$ for some $r \in (0,1)$. But instead of normalizing the values on E_m we will normalize E_{m+n}, thus getting

$$r^{-n}E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-n}E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u. The constant r is called the *renormalization constant*. From the equality above we can define the following

Definition 2.6: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m}E_m(u).$$

Note that the sequence $\{\mathcal{E}_m(u)\}$ is a non-decreasing sequence.

Based on this definition we see that for a harmonic extension $\tilde{u}$ of u we have that

$$\mathcal{E}_{m+1}(\tilde{u}) = r^{-(m+1)}E_{m+1}(\tilde{u}) = r^{-m}E_m(u) = \mathcal{E}_m(u).$$

Extending this to the bilinear form we have, for harmonic extension $\tilde{u}, \tilde{v}$ of u, v that

$$\mathcal{E}_{m+1}(\tilde{u}, \tilde{v}) = \mathcal{E}_m(u,v).$$

This is also true for the bilinear form, if $\tilde{u}$ is the harmonic extension of u and v' is any extension of v then

$$\mathcal{E}_{m+1}(\tilde{u}, v') = \mathcal{E}_m(u,v). \tag{2.1}$$

See [27] for further details.

Now that we have definition 2.6, it is possible to define a harmonic function.

Definition 2.7: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m, given boundary values on V_0.

The harmonic functions will be used extensively throughout this section. We start by finding the renormalization constant for the Vicsek set.

2.1.1 The Renormalization Constant on the Vicsek set

In the previous section we talked about a harmonic function, but without ever constructing one. We will therefore in this section show how to construct a harmonic function for the Vicsek set for any initial boundary values $u(V_0)$.

Given $V_0 = \{q_1, q_2, q_3, q_4, q_5\}$ and $u\colon V_0 \to \mathbb{R}$ we have that

$$E_0(u) = (u(q_1) - u(q_2))^2 + (u(q_1) - u(q_3))^2 + (u(q_1) - u(q_4))^2 + (u(q_1) - u(q_5))^2 .$$

We are going to exploit the symmetry of the Vicsek set. Since all four "extremities" of the Vicsek set are identical we will only look at one of the limbs, since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, for $u' : V_1 to \mathbb{R}$, we see for the extremity between q_1 and q_2 that

$$\begin{aligned} E_1(u') = \Big(u'(q_2) - u'(p_1)\Big)^2 + \Big(u'(p_1) - u'(p_3)\Big)^2 + \Big(u'(p_1) - u'(p_4)\Big)^2 \\ + \Big(u'(p_1) - u'(p_5)\Big)^2 + \Big(u'(p_4) - u'(q_1)\Big)^2 . \end{aligned} \tag{2.2}$$

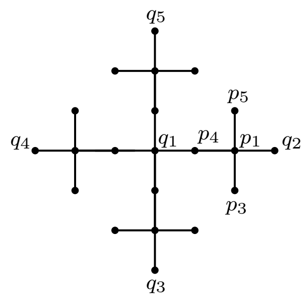

Figure 2.1: Points on the Vicsek set

Next we are going to optimize with regards to $u'(p_i)$ for $i = 0, 1, 2, 4$. Taking the derivative of the right side of 2.2 and solving for 0 we get the following

$$\begin{aligned} 4u'(p_1) &= u'(q_2) + u'(p_3) + u'(p_4) + u'(p_5) \\ 2u'(p_4) &= u'(q_1) + u'(p_1) \\ u'(p_3) &= u'(p_1) \\ u'(p_5) &= u'(p_1). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_1) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_3) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_5) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_2). \end{aligned}$$

Ergo what we have found is, that to minimize the value in the points of the Vicsek set, the value in the next point is going to be 2/3 the value from the closest point and 1/3 the value from the second closest point.

If we calculate the graph energy we find that

$$E_1(u') = \frac{1}{3}\sum_{i=1}^{4}(u'(q_i) - u'(q_1))^2 = \frac{1}{3}E_0(u').$$

Since the Vicsek set is a self-similar set, we have that going from V_1 to V_2 is the same as going from V_m to V_{m+1}. Thus we can simply extend the above calculations to doing it m times, in which case

$$E_{m+1}(u') = \frac{1}{3}E_m(u').$$

We then get our renormalization constant for the Vicsek set to be $r = \frac{1}{3}$. Altogether we find that from definition 2.6 we have that

$$\mathcal{E}_m(u) = 3^m E_m(u).$$

2.2 Kigami Laplacian

With the renormalization constant found, we will return to finding the Kigami Laplacian. Recall from definition 2.6 that we have defined a sequence $\mathcal{E}_m$ for functions on V^*. It will therefore make sense to define the following limit.

Definition 2.8: Let $u\colon V^* \to \mathbb{R}$ then define the limit

$$\mathcal{E}(u,u) = \lim_{m\to\infty} \mathcal{E}_m(u,u)$$

where we set

$$\operatorname{dom}\mathcal{E} = \left\{u \in \ell(V_*) : \mathcal{E}(u) < \infty\right\}, \quad \text{and} \quad \operatorname{dom}_0\mathcal{E} = \left\{u \in \operatorname{dom}\mathcal{E} : u = 0, \text{ on } V_0\right\}.$$

This definition only involves functions defined on V^*, but we find that functions in $\operatorname{dom}\mathcal{E}$ are uniformly continuous and therefore have a unique extension to K. We even have that $\operatorname{dom}\mathcal{E}$ is dense in $C(K)$.

Proposition 2.9: Let $\operatorname{dom}\mathcal{E}$ be defined as in Definition 2.8. Then the following holds

(i) Let $u \in \operatorname{dom}\mathcal{E}$ then u is uniformly continuous on V^*.

(ii) $\operatorname{dom}\mathcal{E}$ is dense in $C(K)$.

Proof. We start with (i). Let $u \in \operatorname{dom}\mathcal{E}$ then $\mathcal{E}_m(u) \le \mathcal{E}(u)$ since $\mathcal{E}_m$ is non-decreasing in m. So for $x, y \in V_m$ if $y \in V_{m,x}$we have that

$$r^{-m}(u(x) - u(y))^2 \le \mathcal{E}_m(u) \le \mathcal{E}(u) \tag{2.3}$$

since $r^{-m}(u(x) - u(y))^2$ is a term in $\mathcal{E}_m(u)$. Thus

$$\left|u(x) - u(y)\right| \le r^{\frac{m}{2}}\mathcal{E}(u)^{\frac{1}{2}}.$$

Now consider a chain of points $x_m, x_{m+1}, \dots, x_{m+k}$ where $x_{m+j} \in V_{m+j}$ and $x_{m+j} \in V_{m+j+1,x_{m+j+1}}$. Then we have

$$\left|u(x_m) - u(x_{m+k})\right| \le r^{\frac{m}{2}}\left(1 + r^{\frac{1}{2}} + \ldots + r^{\frac{k}{2}}\right)\mathcal{E}(u)^{\frac{1}{2}} \le \frac{r^m}{1 - r^{\frac{1}{2}}}\mathcal{E}^{\frac{1}{2}}$$

by using (2.3) and the triangle inequality multiple times. From the geometry of K we see that $x, y \in V^*$ belong to the same or adjacent $(\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m})(K)$. We can then connect x to y by one of such chains, so

$$|u(x) - u(y)| \le \frac{2r^{m/2}}{1 - r^{\frac{1}{2}}} \mathcal{E}(u)^{\frac{1}{2}}$$

giving a bound for all $x, y \in V^*$.

For proof of (ii) we refer to [27]. □

We are now ready to define the Kigami Laplacian. We will denote this with Δ_μ as it depends on the choice of measure μ. In our case we choose μ to be as in Definition 1.13.

Definition 2.10: Let $u \in \operatorname{dom} \mathcal{E}$ and let $f \in C(K)$, then $u \in \operatorname{dom} \Delta_\mu$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u, v) = \int_K f v \, \mathrm{d}\mu, \quad \forall v \in \operatorname{dom}_0 \mathcal{E}.$$

We call Δ_μ the *Kigami Laplacian.*

As mentioned before, we want Δ_μ to be defined as the limit of Δ_m. The constants 3^m and 5^m are not random, but chosen such that the Kigami Laplacian makes sense. But to prove the pointwise limit and actually calculate the constants, we will need a few lemmas and definitions. We will introduce a new notation for the discrete graph Laplacian without its constants. The reason for this will become apparent, since we in future proofs will have that the constants cancels each other out. We start by defining this graph Laplacian followed by a harmonic function which will be used extensively in the following proofs.

Definition 2.11: Let Δ_m be defined as in Definition 2.1. We then define $\widetilde{\Delta}_m$ as

$$\widetilde{\Delta}_m f(p) = 3^{-m} N_p \Delta_m f(p) = \sum_{q \in V_{m,p}} f(q) - f(p).$$

Definition 2.12: Given m and a point $x \in V_m \backslash V_0$ then define the harmonic function $\psi_x^m \colon V^* \to [0, 1]$ that for a $y \in V_m$ is $\psi_x^{(m)}(y) = \delta_{xy}$.

Note that $\phi_x^m \in \operatorname{dom} \mathcal{E}$. Furthermore, it might seem odd that we have only defined $\psi_x^{(m)}$ to have values on V_m. But the function is defined on the whole of V^*, where the values in $K \setminus V_m$ come from the fact that it is a harmonic extension. We even have that function $\psi_x^{(m)}$ can be used to describe $\widetilde{\Delta}_m$.

Proposition 2.13: Let $u \in \operatorname{dom} \mathcal{E}$ and $\Delta_\mu u = f$ then

$$r^{-m} \widetilde{\Delta}_m u(x) = \int_K f \psi_x^{(m)} \, \mathrm{d}\mu$$

Proof. We see that

$$\mathcal{E}_m(u, \psi_x^{(m)}) = r^{-m} \sum_{p \underset{m}{\sim} q} (u(q) - u(p))(\psi_x^{(m)}(p) - \psi_x^{(m)}(q)) = r^{-m} \sum_{q \in V_{m,x}} (u(x) - u(q)) = r^{-m} \widetilde{\Delta}_m u(x)$$

where the second equality stems from the fact that only edges containing x are going to see change in $\psi_x^{(m)}$. By (2.1) we have that

$$r^{-m} \widetilde{\Delta}_m u(x) = \mathcal{E}_m(u, \psi_x^{(m)}) = \mathcal{E}(u, \psi_x^{(m)}) = \int_K f \psi_x^{(m)} \, \mathrm{d}\mu$$

as desired. □

Lemma 2.14: Let $v\colon V^* \to \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{k>0} S^k(x)\right) \cup \left(\bigcup_i \{y_i\}\right)$.

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the value of the points $S^k(x)$ with $k > 0$ depends on the value of x and its neighbours.

So we need to check the points in $\overline{B_x} \setminus B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \to \omega$. By the fact that v is continuous on K it is continuous on $\overline{B_x}$, in which case $\left|v(x_n) - v(\omega)\right| \to 0$ as $n \to \infty$, but $v(x_n) = 0$ for all x_n, so $v(\omega) = 0$ as desired. □

WhatLemma 2.14 states is, if any two points in V_m are neighbours, say x and y, and $v(x) = v(y) = 0$, then the function value at any point between x and y is also zero.

Lemma 2.15: Set $F_m = \operatorname{supp} \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

Further F_m is contained in the open ball $B(x, \varepsilon_m)$ where $\varepsilon_m \to 0$ as $m \to \infty$.

Proof. If we set $\Omega_m = \left\{ z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z} \right\}$ then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B_\alpha} \right)$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have, given the constructed of F_m, that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \varepsilon_m)$ follows from Lemma 2.14 and that $F_m \supset F_{m+1}$. □

We are now ready to prove the pointwise formula.

Theorem 2.16: Let $u \in \operatorname{dom} \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \to \infty} r^{-m} \left(\int_K \psi_x^{(m)} \, \mathrm{d}\mu \right)^{-1} \widetilde{\Delta}_m u(x) \tag{2.4}$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuous function and the right side of (2.4) converges uniformly to a continuous function on $V_* \setminus V_0$. Then $u \in \operatorname{dom} \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \operatorname{dom} \Delta_\mu$ and $\Delta_\mu u = f$, then by Proposition 2.13 we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} \, \mathrm{d}\mu \right)^{-1} \widetilde{\Delta}_m u(x) = \frac{\int_K f \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu}$$

for any $x \in V_m \setminus V_0$. The right side needs to converge uniformly, so let $\varepsilon > 0$. Since $f(x)$ is continuous then by Lemma 2.15 we can choose m large enough such that for all $x, y \in F_m$ we have that $\left|f(x) - f(y)\right| < \varepsilon$, in which case

$$\begin{aligned}
\left| \frac{\int_K f \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} - \frac{\int_K f(x) \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} \right| \\
&= \left| \frac{\int_K \left(f(y) - f(x)\right) \psi_x^{(m)} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} \right| \\
&\leq \frac{\int_{F_m} \left|f(y) - f(x)\right| \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} \\
&\leq \frac{\varepsilon \int_{F_m} \, \mathrm{d}\mu}{\int_K \psi_x^{(m)} \, \mathrm{d}\mu} \\
&= \varepsilon
\end{aligned}$$

in which case

$$\lim_{m\to\infty} r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x) = \lim_{m\to\infty}\frac{\int_K f\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} = f(x) = \Delta_\mu u(x)$$

which proves the first statement.

Next suppose that the expression $r^m\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)$ converges uniformly to a continuous function f. Then for any $v\in\mathrm{dom}_0\,\mathcal{E}$ we have

$$\mathcal{E}_m(u,v) = r^{-m}\sum_{x\underset{m}{\sim}y}\left(u(y)-u(x)\right)\left(v(y)-v(x)\right).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x\in V_m\setminus V_0$ then each $v(x)$ will be of the form $-v(x)\left(u(y)-u(x)\right)$ for each $y\underset{m}{\sim}x$. We therefore get

$$\begin{aligned}\mathcal{E}_m(u,v) &= -r^{-m}\sum_{x\in V_m\setminus v_0} v(x)\left(\sum_{x\underset{m}{\sim}y}\left(u(y)-u(x)\right)\right)\\ &= -\sum_{x\in V_m\setminus V_0} v(x)r^{-m}\widetilde{\Delta}_m u(x)\end{aligned}$$

While it seems we lose many terms by removing $v(y)$, they are instead contained in

$$\sum_{x\underset{m}{\sim}y}\left(u(y)-u(x)\right).$$

Note that the sum is "reset" for every x so we count both the relation $x\sim y$ and $y\sim x$.

Now if we define $f_m(x) = r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)$ then $f_m\to f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned}\mathcal{E}_m(u,v) &= -\sum_{x\in V_m\setminus V_0} v(x)\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)\\ &= -\int_K\left(\sum_{x\in V_m\setminus V_0} v(x)f_m(x)\psi_x^{(m)}\right)\mathrm{d}\mu.\end{aligned}$$

Since the convergence in the next calculation follows from a quite lengthy argument, we will show the calculation and then return to the convergence. We have that

$$\mathcal{E}(u,v) = \lim_{m\to\infty}\mathcal{E}_m(u,v) = \lim_{m\to\infty} -\int_K\left(\sum_{x\in V_m\setminus V_0} v(x)f_m(x)\psi_x^{(m)}\right)\mathrm{d}\mu = -\int_K v(y)f(y)\,\mathrm{d}\mu \quad (2.5)$$

therefore $u\in\mathrm{dom}\Delta_\mu$. Now looking at the convergence in (2.5) we find

$$\begin{aligned}&\left|\sum_{x\in V_m\setminus V_0} v(x)f_m(x)\psi_x^{(m)}(y) - v(y)f(y)\right|\\ &\qquad\le\left|\sum_{x\in V_m\setminus V_0}\left(v(x)f_m(x)-v(y)f(y)\right)\psi_x^{(m)}(y)\right| + \left|\sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y)\right|.\end{aligned}$$

Looking at each term individually we first see that

$$\left|\sum_{x\in V_m\setminus V_0}\left(v(x)f_m(x)-v(y)f(y)\right)\psi_x^{(m)(y)}\right|$$
$$\leq \sum_{x\in V_m\setminus V_0}\Big(\left|v(x)f_m(x)-v(y)f_m(y)\right|+\left|v(y)f_m(y)-v(y)f(y)\right|\Big)\,\psi_x^{(m)}(y)$$

We will again look at each term starting with $\left|v(y)f_m(y)-v(y)f(y)\right|$. Here

$$\left|v(y)f_m(y)-v(y)f(y)\right|=\left|v(y)\right|\left|f_m(y)-f(y)\right|$$

and since v is continuous on K, which is compact, there is a constant C_1 so $\left|v(y)\right|\leq C_1$ for all $x\in K$. Since f_m converges uniformly to f we can choose m_1 such that $\left|f_{m_1}(y)-f(y)\right|\leq\frac{\varepsilon}{C_1}$ and the convergence of that term follows.

Next is $\left|v(x)f_m(x)-v(y)f_m(y)\right|$. Here we find

$$\left|v(x)f_m(x)-v(y)f_m(y)\right|\leq\left|v(x)\right|\left|f_m(x)-f(y)\right|+\left|f_m(y)\right|\left|v(x)-v(y)\right|$$

First $\left|f_m(y)\right|\left|v(x)-v(y)\right|$. Here $\left|f_m(y)\right|\leq C_2$ and by Lemma 2.15 and continuity of v we can choose m_2 so $\left|v(x)-v(y)\right|\leq\frac{\varepsilon}{C_2}$ for all $x\in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2}.

Second we examine $\left|v(x)\right|\left|f_m(x)-f(y)\right|$. Again $\left|v(x)\right|\leq C_1$, but

$$\left|f_m(x)-f(y)\right|\leq\left|f_m(x)-f(x)\right|+\left|f(x)-f_m(y)\right|\leq\left|f_m(x)-f(x)\right|+\left|f(x)-f(y)\right|+\left|f(y)-f_m(y)\right|$$

First choose m_3 so $\left|f_{m_3}(x)-f(x)\right|\leq\frac{\varepsilon}{3C_1}$. Next by continuity of f choose m_4 so $\left|f(x)-f(y)\right|\leq\frac{\varepsilon}{3C_1}$ for all $x\in F_{m_4}$. Lastly choose m_5 so $\left|f(y)-f_{m_5}(y)\right|\leq\frac{\varepsilon}{3C_1}$.

So we now have for $m\geq\max\{m_1,m_2,\dots,m_5\}$ that

$$\left|\sum_{x\in V_m\setminus V_0}\left(v(x)f_m(x)-v(y)f(y)\right)\psi_x^{(m)}(y)\right|\leq\varepsilon\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(y)\leq\varepsilon$$

We are going to look at

$$\left|\sum_{x\in V_m\setminus V_0}v(y)f(y)\psi_x^{(m)}(y)-v(y)f(y)\right|=\left|v(y)f(y)\right|\left|\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(y)-1\right|$$

and show that

$$\left|\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(y)-1\right|=0,\quad\text{as } m\to\infty. \tag{2.6}$$

If $y\in V^*$, then choose m_6 so $y\in V_{m_6}$ and (2.6) holds.

Next assume $y\in K\setminus V^*$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n\to y$ as $n\to\infty$, and since $\psi_x^{(m)}$ is continuous on K then $\psi_x^{(m)}(x_n)\to\psi_x^{(m)}(y)$ as $n\to\infty$. This gives us the following

$$\left|\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(y)-1\right|\leq\left|\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(y)-\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(x_n)\right|+\left|\sum_{x\in V_m\setminus V_0}\psi_x^{(m)}(x_n)-1\right|.$$

Now choose an n_1 so $\left|\psi_x^{(m)}(x_{n_1})-\psi_x^{(m)}(y)\right|\leq\varepsilon$ and an m_7 so $x_{n_1}\in V_{m_7}$. Both can then be bounded by $m\geq\max\{n_1,m_7\}$. Thus we have the convergence of (2.5). □

Lastly we calculate the constant $\left(\int_K \psi_x^{(m)}(y)\ \mathrm{d}\mu(y)\right)^{-1}$.

Lemma 2.17:

$$\int_K \psi_x^m(y)\ \mathrm{d}\mu(y) = \begin{cases} 5^{-m} & \text{if } \#V_{m,p} = 4 \\ 2 \cdot 5^{-(m+1)} & \text{if } \#V_{m,p} = 2 \\ 5^{-(m+1)} & \text{if } \#V_{m,p} = 1 \end{cases}$$

Proof. Recall from eq. (1.3) that we can write a point x's "address" with the infinite sequence $K_{i_1,i_2,\ldots} = (\mathcal{S}_{i_1} \circ \mathcal{S}_{i_2} \circ \ldots)(K)$. These sequences are not necessarily unique and we will have three cases. The first two cases are where x has a unique sequence and $\#V_{m,p} \in \{1,4\}$. The last case is where x has two sequences and $\#V_{m,p} = 2$. If we start with two sequences, then we have that $x \in K_{i_1,\ldots,i_m} \cap K_{i_1,\ldots,i_{m'}}$ in which case from Lemma 2.14 it follows that $\operatorname{supp} \psi_x^m = K_{i_1,\ldots,i_{m+1}} \cup K_{i_1,\ldots,i_{m'+1}}$, for an appropriate choice of i_{m+1} and $i_{m'+1}$. Now if $x \in K_{i_1,\ldots,i_m}$ and $\#V_{m,p} = 4$ we have that $\operatorname{supp} \psi_x^m = K_{i_1,\ldots,i_m}$, but if $\#V_{m,p} = 1$ then $\operatorname{supp} \psi_x^m = K_{i_1,\ldots,i_{m+1}}$ again for an appropriate i_{m+1}.

If we now look at an $\alpha \in \operatorname{supp} \psi_x^m$ it is going to have a sequence of the form $K_{i_1,\ldots i_m,\alpha_{j_1},\alpha_{j_2}\ldots}$ or $K_{i_1,\ldots i_k,\alpha_{j_1},\alpha_{j_2}\ldots}$ where $k \in \{m+1, m'+1\}$. We therefore have

$$\int_K \psi_x^{(m)}(y)\ \mathrm{d}\mu(y) = \int_{\operatorname{supp}\psi_x^m} \psi_x^{(m)}(y)\ \mathrm{d}\mu(y) = \sum_{\alpha\in\operatorname{supp}\psi_x^m} \int_{\{\alpha\}} \psi_x^m(y)\ \mathrm{d}\mu(y)$$

If x has a unique sequence and $\#V_{m,p} = 4$ then

$$\begin{aligned}\sum_{\alpha\in\operatorname{supp}\psi_x^m} \int_{\{\alpha\}} \psi_x^m(y)\ \mathrm{d}\mu(y) &= \sum_{\alpha\in\operatorname{supp}\psi_x^m} \mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \ldots)(K)\right) \\ &= \mu\left(\bigcup_{n\geq 0} (\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m} \circ \mathcal{S}_{\alpha_{j_n}} \circ \ldots)(K)\right) = \mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m})(K)\right) = 5^{-m}.\end{aligned}$$

By the almost exact same calculation we find that if x has a unique sequence and $\#V_{m,p} = 1$ then

$$\sum_{\alpha\in\operatorname{supp}\psi_x^m} \int_{\{\alpha\}} \psi_x^m(y)\ \mathrm{d}\mu(y) = 5^{-(m+1)}.$$

Now if x does not have a unique sequence then

$$\begin{aligned}\sum_{\alpha\in\operatorname{supp}\psi_x^m} \int_{\{\alpha\}} \psi_x^m(y)\ \mathrm{d}\mu(y) &= \sum_{\alpha\in\operatorname{supp}\psi_x^m} \Bigg[\mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m+1} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \ldots)(K)\right) \\ &\qquad + \mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i'_m+1} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \ldots)(K)\right)\Bigg] \\ &= \mu\left(\bigcup_{n\geq 0} (\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m+1} \circ \mathcal{S}_{\alpha_{j_n}} \circ \ldots)(K)\right) \\ &\qquad + \mu\left(\bigcup_{n\geq 0} (\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i'_m+1} \circ \mathcal{S}_{\alpha_{j_n}} \circ \ldots)(K)\right) \\ &= \mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_m+1})(K)\right) + \mu\left((\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i'_m+1})(K)\right) \\ &= 2 \cdot 5^{-(m+1)}\end{aligned}$$

and the result follows. □

2.3 Dirichlet Laplacian

We will now construct the Dirichlet Laplacian. Since the Dirichlet Laplacian is a densely defined operator we will therefore start by recalling some definitions for a densely defined operator on a Hilbert space. We will also need Dirichlet forms, thus we will also be defining them.

Definition 2.18: Let $\mathcal{H}$ be a Hilbert space and let $A\colon \operatorname{dom} A \to \mathcal{H}$ be a densely defined operator. Then we say the following about A:

(i) A is said to be symmetric if for $f, g \in \operatorname{dom} A$

$$\langle Af, g\rangle = \langle f, Ag\rangle .$$

(ii) A is said to be non-negative (resp. non-positive) if for $f \in \operatorname{dom} A$

$$\langle Af, f\rangle \geq 0, \quad (\text{resp.}\ \langle Af, f\rangle \leq 0)$$

(iii) A is said to be self-adjoint if $\operatorname{dom} A^* = \operatorname{dom} A$ where A^* is the adjoint of A on the domain

$$\operatorname{dom} A^* = \Big\{ f \in \mathcal{H} : \exists c(f) \geq 0, \forall g \in \mathcal{D}(A), |\langle f, Ag\rangle| \leq c(f)\|g\| \Big\} .$$

and $c(f)$ denotes that the constant may depend on f. We can further define A^* by the formula

$$\langle A^* f, g\rangle = \langle f, Ag\rangle , \quad \text{for } f \in \operatorname{dom} A^*, g \in \operatorname{dom} A.$$

Definition 2.19: A *quadratic form* $\mathcal{E}$ on $\mathcal{H}$ is a non-negative, definite, symmetric bilinear form from $\operatorname{dom}\mathcal{E} \times \operatorname{dom}\mathcal{E} \to \mathbb{R}$, where $\operatorname{dom}\mathcal{E}$ is a dense subspace of $\mathcal{H}$.

(i) A quadratic form is said to be closed if $\operatorname{dom}\mathcal{E}$ equipped with the norm

$$\|f\|^2_{\operatorname{dom}\mathcal{E}} = \|f\|^2 + \mathcal{E}(f,f)$$

is a Hilbert space.

(ii) A quadratic form $\mathcal{E}$ on $\mathcal{H}$ is said to be closable if it admits a closed extension. That is, there exists a closed quadratic form $\overline{\mathcal{E}}$ such $\operatorname{dom}\mathcal{E} \subset \operatorname{dom}\overline{\mathcal{E}}$ and $\overline{\mathcal{E}}$ coincides with $\mathcal{E}$ on $\operatorname{dom}\mathcal{E} \times \operatorname{dom}\mathcal{E}$.

(iii) We further call a quadratic form $\mathcal{E}$ a Dirichlet form if it has the following property. If given $u, v \in \operatorname{dom}\mathcal{E}$ so that for almost every $x, y \in K$, we have

$$\left|v(x) - v(y)\right| \leq \left|u(x) - u(y)\right|, \quad \text{and} \quad \left|v(x)\right|, \leq \left|u(x)\right|$$

then

$$\mathcal{E}(v,v) \leq \mathcal{E}(u,u).$$

Now recall from definition 2.8 that

$$\mathcal{E}(f,f) = \lim_{m\to\infty} \mathcal{E}_m(f,f)$$

with domain $\operatorname{dom}_0 \mathcal{E}$ where

$$\mathcal{E}_m(f,f) = \sum_{x \underset{m}{\sim} y} (f(x) - f(y))^2.$$

By [14, Theorem 4.1] we have that $(\mathcal{E}, \operatorname{dom}_0 \mathcal{E})$ is a Dirichlet form on $L^2(K, \mu)$. By [3, Lemma 1.3.3] we have that

$$\mathcal{E}(f, f) = -\int_K \Delta f g \, \mathrm{d}\mu$$

where Δ a unique densely defined self-adjoint operator. The operator Δ is called the generator of $\mathcal{E}$. Moreover we have that Δ is an extension of the Kigami Laplacian Δ_μ. The operator Δ with domain $\operatorname{dom} \Delta$ is referred to as the Dirichlet Laplaican.

Notice that the Kigami Laplacian Δ_μ was not used in the construction of the Dirichlet Laplacian Δ, but that we used the limit of the renormalized graph energy from Definition 2.8. This stems from the fact that there are two ways to define Δ, one through Dirichlet forms and one trough Δ_μ. We have chosen to define Δ through Dirichlet forms, but defining Δ through Δ_μ follows a similarly condensed argument. One would prove that $\Delta_\mu \colon \operatorname{dom} \Delta_\mu \to L^2(K, \mu)$ is a densely defined non-positive symmetric operator. It then follows from [3, Lemma 1.3.5] that the quadratic form $\mathcal{E}(f, g) = -\langle f, \Delta_\mu g \rangle$ for $f, g \in \operatorname{dom} \Delta_\mu$ is closable and the generator of $\mathcal{E}$ is the self-adjoint extension Δ.

2.4 Discrete Fractional Laplacian

Recalling Remark 2.2 we consider the eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \ldots \leq \lambda_{N_m}^m$, each being repeated accordingly to multiplicity, of $-\Delta_m$. Next let $(\Phi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions with respect to the measure μ_m. This leads us to defining the following.

Definition 2.20: Let $s \geq 0$, we then define the *discrete Riesz kernel* on V_m as

$$G_s^m(x, y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m.$$

By construction we can see that the matrix $(G_s^m(x, y))_{x, y \in V_m}$ is symmetric. It is also positive semidefinite, which follows from

$$\begin{aligned}
\sum_{i,j=1}^{N_m} a_i a_j G_s^m(x_i, x_j) &= \sum_{i,j=1}^{N_m} a_i a_j \sum_{k=1}^{N_m} (\lambda_k^m)^{-s} \phi_k^m(x) \phi_k^m(y) \\
&= \sum_{k=1}^{N_m} \lambda_k^{-s} \left(\sum_{i=1}^{N_m} \phi_k(x_i)^2 a_i^2 + \sum_{i,j=1, i \neq j}^{N_m} \phi_k(x_i) \phi_k(x_j) a_i a_j \right) \\
&= \sum_{k=1}^{N_m} \lambda_k^{-s} \left(\sum_{i=1}^{N_m} \phi_k(x_i) a_i \right)^2 \geq 0
\end{aligned}$$

Thus it can be the covariance matrix of a Gaussian vector. The Riesz kernel will allow us to define the discrete fractional Laplacian.

Definition 2.21: For $f \in \ell(V_m)$ and $s \geq 0$, the *discrete fractional Laplacian* $(-\Delta_m)^s \colon L^2(V_m, \mu_m) \to L^2(V_m, \mu_m)$ is defined by

$$(-\Delta_m)^{-s} f(x) = \frac{1}{a_m} \sum_{y \in V_m} G_s^m(x, y) f(y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y).$$

Remark: Note that $\frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) = \left\langle \phi_j^m, f \right\rangle_{L^2(V_m, \mu_m)}$ and therefore that $(-\Delta_m)^{-s} f(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \left\langle \phi_j^m, f \right\rangle_{L^2(V_m, \mu_m)}$.

Having stated the discrete fractional Laplacian we move on to proving some of its useful properties.

Proposition 2.22: Let $(-\Delta_m)^{-s}$ be defined as in Definition 2.21 then the follow holds

(i) $(-\Delta_m)^{-s}$ is a uniformly bounded operator

(ii) $(-\Delta_m)^{-s}$ is a symmetric operator.

Proof. Here $\|\cdot\| = \|\cdot\|_{L^2(V_m,\mu_m)}$ and $\langle\cdot,\cdot\rangle = \langle\cdot,\cdot\rangle_{L^2(V_m,\mu_m)}$. We start with (i). We have

$$\begin{aligned}
\left\|(-\Delta_m)^{-s}f(x)\right\| &\leq \left\|\sum_{i=1}^{N_m}(\lambda_i^m)^{-s}\phi_i^m(x)\frac{1}{a_m}\sum_{y\in V_m}\phi_i^m(y)f(y).\right\| \\
&\leq \sum_{i=1}^{N_m}(\lambda_i^m)^{-s}\frac{1}{a_m}\left\|\phi_i^m(x)\sum_{y\in V_m}\phi_i^m(y)f(y)\right\| \\
&\leq (\lambda_1^m)^{-s}\|f\|
\end{aligned}$$

where the last inequality comes from the fact that $0<\lambda_1<\lambda_2<\ldots$. The uniform bound comes from the fact that $\inf_m \lambda_1^m > 0$. This follows from [14, Lemma 5.2], since if $\inf_m \lambda_1^m \leq 0$ then the series in the Lemma would not converge.

Moving on to (ii), we have that

$$\begin{aligned}
\left\langle f,(-\Delta_m)^{-s}g\right\rangle &= \int_{V_m} f(-\Delta_m)^s g \,\mathrm{d}\mu_m(x) \\
&= \int_{V_m} f(x)\cdot\left(\sum_{ij=1}^{N_m}(\lambda_i^m)^{-s}\phi_j^m(x)\frac{1}{a_m}\sum_{y\in V_m}\phi_i^m(y)g(y)\right)\,\mathrm{d}\mu_(x) \\
&= \sum_{j=1}^{N_m}(\lambda_i^m)^{-s}\frac{1}{a_m}\sum_{y\in V_m}\phi_j^m(y)g(y)\int_{V_m}\left(\sum_{i=1}^{N_m}\phi_i^m(x)\left\langle\phi_i^m,f\right\rangle\right)\phi_j^m(x)\,\mathrm{d}\mu_m(x) \\
&= \sum_{j=1}^{N_m}(\lambda_i^m)^{-s}\frac{1}{a_m}\sum_{y\in V_m}\phi_j^m(y)g(y)\int_{V_m}\phi_j^m(x)\left\langle\phi_j^m,f\right\rangle\phi_j^m(x)\,\mathrm{d}\mu_m(x) \\
&= \sum_{j=1}^{N_m}\Bigg[\int_{V_m}\left((\lambda_j^m)^{-s}\phi_j(x)\left\langle\phi_j^m,f\right\rangle\right)\phi_j^m(x)\left\langle\phi_j^m,g\right\rangle\,\mathrm{d}\mu_m(x) \\
&\qquad + \int_{V_m}\left((\lambda_j^m)^{-s}\phi_j^m(x)\left\langle\phi_j^m,f\right\rangle\right)\left(\sum_{i=1,i\neq j}^{N_m}\phi_i^m(x)\left\langle\phi_i^m,g\right\rangle\right)\,\mathrm{d}\mu_m(x)\Bigg] \\
&= \int_{V_m}(-\Delta_m)^{-s}fg\,\mathrm{d}\mu_m(x) \\
&= \left\langle(-\Delta_m)^{-s}f,g\right\rangle
\end{aligned}$$

using that $f=\sum_{i=1}^{N_m}\phi_i\langle\phi_i,f\rangle$ and $\int_{V_m}\phi_i\phi_j\,\mathrm{d}\mu_m$ is 1 if $j=i$ and 0 if $i\neq j$. □

2.5 Semigroups and Heat kernels

When constructing the fractional Laplacian in Section 2.6, we will use that $-\Delta$ generates a semigroup which admits a heat kernel. We will therefore first define a semigroup as well as showing that $-\Delta$ generates a semigroup. We will also define the heat kernel.

Definition 2.23: Let $\mathcal{H}$ be a Hilbert space. A *strongly continuous self-adjoint contraction semigroup* is a family of self-adjoint operators $(P_t)_{t\geq 0} \colon \mathcal{H} \to \mathcal{H}$

(i) For $s, t \geq 0$, $P_t \circ P_s = P_{s+t}$ (semigroup property).

(ii) For every $f \in \mathcal{H}, \lim_{t\to 0} P_t f = f$ (strong continuity).

(iii) For every $f \in \mathcal{H}$ and $t \geq 0, \|P_t f\| \leq \|f\|$ (contraction property).

Theorem 2.24 is vital as it will show that an operator can generate a semigroup.

Theorem 2.24: If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{-tA}$.

Proof. Let A be given as stated. From the spectral theorem [3, Theorem 1.1.2] there is a measure space (Ω, ν), a unitary map $U \colon L^2(\Omega, \nu) \to \mathcal{H}$ and a non negative real valued measurable function λ on Ω such that

$$U^{-1}AUf(x) = -\lambda(x)f(x)$$

for $x \in \Omega, Uf \in \operatorname{dom} A$. Then from the functional calculus [3, Theorem 1.1.3] we define $P_t \colon \mathcal{H} \to \mathcal{H}$ such that

$$U^{-1}P_tUf(x) = e^{-t\lambda(x)}f(x).$$

Note that since U is a unitary map, it is a linear bijection that satisfies for all $x, y \in L^2(\Omega, \nu)$ that $\langle Ux, Uy\rangle_{\mathcal{H}} = \langle x, y\rangle_{L^2(\Omega,\nu)}$.

It is enough to show that $Q_t = U^{-1}P_tU$ is self-adjoint, since then P_t will be self-adjoint. This can be proven quite easily. If we have the functions $f, g \in \mathcal{H}$ and $\psi, \phi \in L^2(\Omega, \nu)$ then P_t is symmetric since

$$\begin{aligned}\langle Af, g\rangle_{\mathcal{H}} = \langle AU\psi, U\phi\rangle_{\mathcal{H}} &= \left\langle UU^{-1}UAU\psi, UU^{-1}U\phi\right\rangle_{\mathcal{H}} = \left\langle U^{-1}AU\psi, \phi\right\rangle_{L^2(\Omega,\nu)} \\ &= \left\langle \psi, U^{-1}AU\phi\right\rangle_{L^2(\Omega,\nu)} = \left\langle U\psi, UU^{-1}AU\phi\right\rangle_{\mathcal{H}} = \langle f, Ag\rangle_{\mathcal{H}}.\end{aligned}$$

That $\operatorname{dom} P_t^* = \operatorname{dom} P_t$ reasons from the same argument as in (2.7), which will be given later in the proof.

We will prove that $Q_t = U^{-1}P_tU$ with domain $\operatorname{dom} Q_t$ is a self-adjoint strongly continuous contraction semigroup on $L^2(\Omega, \nu)$ with generator A. Starting with A being a generator. We see that for $f \in \operatorname{dom} A$

$$\begin{aligned}\lim_{t\to 0}\left\|\frac{Q_tf - f}{t} - Af\right\| &= \lim_{t\to 0}\sqrt{\left\langle \frac{Q_tf - f}{t} - Af, \frac{Q_tf - f}{t} - Af\right\rangle} \\ &= \lim_{t\to 0}\sqrt{\left\langle \frac{Q_tf - f}{t}, \frac{Q_tf - f}{t} - Af\right\rangle - \left\langle Af, \frac{Q_tf - f}{t} - Af\right\rangle} \\ &\leq \lim_{t\to 0}\sqrt{\left\langle \frac{Q_tf - f}{t}, \frac{Q_tf - f}{t} - Af\right\rangle} \\ &= \lim_{t\to 0}\sqrt{\int_\Omega \left(\frac{Q_tf - f}{t}\right)\left(\frac{Q_tf - f}{t} - Af\right)\, \mathrm{d}\nu} \\ &= \lim_{t\to 0}\sqrt{\int_\Omega \left(\frac{(e^{-\lambda t} - 1)f}{t}\right)\left(\frac{Q_tf - f}{t} - Af\right)} \\ &= 0\end{aligned}$$

where the convergence comes from the fact that $e^{-\lambda t}$ converges to 1 much faster than $1/t$ diverges to ∞. Thus A is a generator. Next we have that Q_t is self-adjoint, for which we start with symmetry

$$\langle Q_t f, g\rangle = \int_\Omega Q_t fg \ \mathrm{d}\nu = \int_\Omega e^{-\lambda t} fg \ \mathrm{d}\nu = \langle f, Q_t g\rangle .$$

Furthermore $\operatorname{dom} Q_t$ is dense in $L^2(\Omega,\nu)$ since if $f_n \to f$ in $L^2(\Omega,\nu)$ and $Q_t f_n \to g$ then

$$\lim_{n\to\infty} Q_t f_n = \lim_{n\to\infty} e^{-\lambda t} f_n = e^{-\lambda t} f = Q_t f.$$

Now showing that $\operatorname{dom} Q_t^* = \operatorname{dom} Q_t$. Since Q_t is symmetric we have from the Cauchy-Schwartz inequality that

$$\left|\langle f, Q_t g\rangle\right| = \left|\langle Q_t f, g\rangle\right| \leq \|Q_t f\| \|g\| . \tag{2.7}$$

Thus we have Q_t is self-adjoint.

To show the semigroup properties from Definition 2.23, we have (i)

$$Q_{t+s} f = e^{-\lambda(t+s)} f = e^{-\lambda t} e^{-\lambda s} f = Q_t Q_s f.$$

Further for (ii) we have

$$\lim_{t\to\infty} Q_t f = \lim_{t\to\infty} e^{-t\lambda} f = f$$

and lastly (iii)

$$\|Q_t f\| = \int_\Omega (e^{-\lambda t} f)^2 \ \mathrm{d}\nu \leq \int_\Omega f^2 \ \mathrm{d}\nu = \|f\| .$$

Thus we have that Q_t is a strongly continuous self-adjoint contraction semigroup on $L^2(\Omega,\nu)$ with generator A. □

The next corollary follows quite easily from Theorem 2.24.

Corollary 2.25: P_t is a linear semigroup.

Proof. Let $f, g \in \mathcal{H}$ and α be some constant. Then

$$\begin{aligned} U^{-1} P_t U\left(\alpha f(x) + g(x)\right) &= e^{-t\lambda(x)}\left(\alpha f(x) + g(x)\right) \\ &= \alpha e^{-t\lambda(x)} f(x) + e^{-t\lambda(x)} g(x) = \alpha U^{-1} P_t U f(x) + U^{-1} P_t U g(x) \end{aligned}$$

□

With semigroups out of the way we can define the heat kernel.

Definition 2.26: Let $(P_t)_{t\geq 0}$ be a strongly continuous self-adjoint contraction semigroup on $L^2(X,\mu)$. We say that the semigroup $(P_t)_{t\geq 0}$ *admits a heat kernel* if the heat kernel measure has a density with respect to μ, i.e. there exists a measurable function $p\colon \mathbb{R}_{>0} \times X \times X \to \mathbb{R}_{\geq 0}$ such that for every $t > 0$, almost everywhere we have

$$P_t f(x) = \int_K p_t(x,y) f(y) \ \mathrm{d}\mu(y)$$

with $x, y \in X$, $f \in L^\infty(X,\mu)$.

Remark 2.27: The discrete Laplacian Δ_m also generates a semigroup on V_m. We denote this by $\{P_t^m\}_{t\geq 0}$ and it is given by $P_t^m = e^{t\Delta_m}$.

2.6 Fractional Laplacian

Before we can define the continuous fractional Laplacian, we need to define the eigenvalues and eigenfunctions for $-\Delta$.

Recall that the Laplacian Δ with domain $\mathcal{D}(\Delta)$ is a densely defined self-adjoint operator. Therefore by Theorem 2.24 it is the generator of a strongly continuous Markov semigroup $\{P_t\}_{t\geq 0}$ on $L^2(K,\mu)$.

From [23, Theorem 8.4] this semigroup admits a heat kernel $p_t(x,y), t>0, x,y\in K$, with respect to the normalized Hausdorff measure μ. The heat kernel $p_t(x,y)$ is called the Dirichlet heat kernel on K. Further from [4, section 2.3] this heat kernel satisfies that for some $c_1, c_2 \in (0,\infty)$

$$p_t(x,y) \leq c_1 t^{-\frac{d_h}{d_w}} \exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right) \tag{2.8}$$

for every $x,y\in K\times K$ and $t\in(0,\infty)$ where d_h is the Hausdorff dimension and d_w is the walk dimension.

The Dirichlet heat kernel is quite central in the definition of the continuous fractional Laplacian, as its bounds and definition will be used quite a bit throughout the proofs of this next section. One of the more important properties of the heat kernel is, that it can be described as a sum of eigenvalues and eigenfunctions of Δ. The following theorem gives us the existence of the eigenvalues, functions and the sum representation of the Dirichlet heat kernel. This theorem will not be proved, but a detailed proof can be found in [11, Theorem 2.2].

Theorem 2.28: There exists an orthonormal basis $\{\phi_i\}_{i=1}^{\infty}$ of $L^2(K,\mu)$ with ϕ_j being eigenfunctions of $-\Delta$. If λ_i is the eigenvalue belonging to ϕ_i, then $\lambda_i > 0$ and $\lim_{i\to\infty}\lambda_i = \infty$. Moreover

$$p_t(x,y) = \sum_{j=1}^{\infty} e^{-\lambda_j t}\phi_j(x)\phi_j(y)$$

where the convergence is uniform for all $t\in(0,\infty)$ and $x,y\in K\times K$

The eigenvalues are not only used to represent the heat kernel but will also be used to represent the fractional Laplacian, and the following lemma gives us an idea of what parameter value gives convergence.

Lemma 2.29: Let $0<\lambda_1\leq\lambda_2<\dots$ be the eigenvalues of $-\Delta$

$$\sum_{j=1}^{\infty}\frac{1}{\lambda_j^{2s}} < \infty \iff s > \frac{d_h}{2d_w}.$$

Proof. This follows from [20, Lemma 5.1.3] which states that there exists constants $c_1, c_2 > 0$ such that

$$c_1\cdot n^{2/d_s} \leq \lambda_n \leq c_2\cdot n^{2/d_s}. \tag{2.9}$$

Assume $s > \frac{d_h}{2d_w}$, then by (2.9)

$$\sum_{j=1}^{\infty}\frac{1}{\lambda_j^{2s}} \leq \sum_{j=1}^{\infty}\frac{1}{(c_1 n^{2/ds})^{2s}} < \infty$$

since

$$\frac{4s}{d_s} > \frac{\frac{2d_h}{d_w}}{\frac{2d_h}{d_w}} = 1.$$

Hence the series is convergent.

Assume now that

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty.$$

Then again by (2.9) we have that

$$\sum_{j=1}^{\infty} \frac{1}{(c_2 n^{2/d_s})^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty$$

and the left sum only converges if $4s/d_s > 1$ in which case $s > d_s/4 = d_h/2d_w$ as desired. □

Thus we are ready to define the fractional Laplacian.

Definition 2.30: Let $s \geq 0$, for $f \in L^2(K, \mu)$ the*fractional Laplacian* $(-\Delta)^{-s}$ is defined as

$$(-\Delta)^{-s} f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, \mathrm{d}\mu(y).$$

Remark: Once again we notice that $\langle \phi_i, f \rangle = \int_K \phi_i f \, \mathrm{d}\mu$, and can rewrite $(-\Delta)^{-s} f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle$.

Now that we have the fractional Laplacian defined, we will show some of its very useful properties.

Proposition 2.31: Let $(-\Delta)^{-s}$ be defined as in Definition 2.30 then:

(i) $(-\Delta)^{-s}$ is a linear operator.

(ii) If ϕ_n be an orthonormal eigenfunction, then $(-\Delta)^{-s} \phi_n = \frac{1}{\lambda_n^{-s}} \phi_n$.

(iii) $(-\Delta)^{-s}$ is a bounded operator.

(iv) $(-\Delta)^{-s}$ is a symmetric operator.

Proof. We begin with the proof of (i).
Let $f, g \in L^2(K, \mu)$ and let α be some constant. Then by definition of $(-\Delta)^{-s}$ we have

$$\begin{aligned}
(-\Delta)^{-s}(\alpha f + g) &= \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y)(\alpha f(y) + g(y)) \, \mathrm{d}\mu(y) \\
&= \alpha \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, \mathrm{d}\mu(y) + \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) g(y) \, \mathrm{d}\mu(y) \\
&= \alpha (-\Delta)^{-s} f + (-\Delta)^{-s} g
\end{aligned}$$

Moving on to (ii). We note that

$$\int_K \phi_j \phi_i \, \mathrm{d}\mu = \delta_{ij}$$

where δ_{ij} is the Kronecker delta. We then have for any eigenfunction ϕ_n that

$$(-\Delta)^{-s} \phi_n = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) \phi_n/y) \, \mathrm{d}\mu(y) = \frac{1}{\lambda_n^{-s}} \phi_n.$$

Next we have (iii). This follows from

$$\begin{aligned}
\left\|(-\Delta)^{-s}f\right\|^2 &= \left\|\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\right\|^2\\
&= \left\langle\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle,\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\right\rangle\\
&= \sum_{j}^{\infty}\left|\frac{1}{\lambda_j^s}\left\langle\phi_j,f\right\rangle\right|^2\left\langle\phi_j,\phi_j\right\rangle\\
&\leq \left(\frac{1}{\lambda_1^s}\right)^2\sum_{j=1}^{\infty}\left|\left\langle\phi_j,f\right\rangle\right|^2\\
&= \left(\frac{1}{\lambda_1^s}\right)^2\|f\|^2
\end{aligned}$$

in which case we have our bound. Finally we will show (iv)

$$\begin{aligned}
\left\langle(-\Delta)^{-s}f,g\right\rangle &= \int_K\left(\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\right)g\ \mathrm{d}\mu\\
&= \int_K\left(\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\right)\left(\sum_{i=1}^{\infty}\phi_i\left\langle\phi_i,g\right\rangle\right)\ \mathrm{d}\mu\\
&= \sum_{j=1}^{\infty}\int_K\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\left(\sum_{i=1}^{\infty}\phi_i\left\langle\phi_i,g\right\rangle\right)\ \mathrm{d}\mu\\
&= \sum_{j=1}^{\infty}\int_K\frac{1}{\lambda_j^s}\phi_j^2\left\langle\phi_j,f\right\rangle\left\langle\phi_j,g\right\rangle\ \mathrm{d}\mu\\
&= \sum_{j=1}^{\infty}\left[\int_K\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,f\right\rangle\phi_j\left\langle\phi_j,g\right\rangle\ \mathrm{d}\mu+\int_K\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,g\right\rangle\left(\sum_{i=1,i\neq j}^{\infty}\phi_i\left\langle\phi_i,f\right\rangle\right)\ \mathrm{d}\mu\right]\\
&= \sum_{j=1}^{\infty}\int_K\left(\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,g\right\rangle\right)\left(\sum_{i=1}^{\infty}\phi_i\left\langle\phi_i,f\right\rangle\right)\ \mathrm{d}\mu\\
&= \int_K\left(\sum_{j=1}^{\infty}\frac{1}{\lambda_j^s}\phi_j\left\langle\phi_j,g\right\rangle\right)f\ \mathrm{d}\mu\\
&= \left\langle f,(-\Delta)^s g)\right\rangle
\end{aligned}$$

as desired. □

While the definition of the fractional Laplacian is not the most complicated, we are able to represent $(-\Delta)^{-}$ in another way, namely with the Riesz kernel.

Definition 2.32: For parameter $s \geq 0$ we define the *fractional Riesz kernel* G_s by

$$G_s(x,y)=\frac{1}{\Gamma(s)}\int_0^{\infty}t^{s-1}p_t(x,y)\ \mathrm{d}t,\quad x,y\in K,x\neq y.$$

Remark: Note that it follows from (2.8) that $G_s(x,y)$ actually converges for every $s \geq 0$ where $x \neq y$.

The Riesz kernel allows us to use the estimates of the heat kernel $p_t(x,y)$ in conjunction with $(-\Delta)^{-s}$ as we shall shortly see. We will start by showing some of the bounds of the fractional Riesz kernel.

Proposition 2.33: Let $G_s(x,y)$ be defined as in Definition 2.32 then the following bounds hold

(i) If $s \in [0, d_h/d_w)$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x,y) \leq \frac{C}{d(x,y)^{d_h - s d_w}}.$$

(ii) If $s = d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x,y) \leq C \Big| \ln\left(d(x,y)\right) \Big|.$$

(iii) If $s > d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K$

$$G_s(x,y) \leq C.$$

Before proving Proposition 2.33 we will quickly state and prove a lemma that is needed.

Lemma 2.34: There exists a constant $C > 0$ such that for every $x, y \in K$ and $t \geq 1$,

$$|p_t(x,y)| \leq C e^{-\lambda_1 t}$$

where $\lambda_1 > 0$ is the first non-zero eigenvalue of $-\Delta$.

Proof. We have from Theorem 2.28 that $p_t(x,y)$ has the expansion

$$p_t(x,y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y). \tag{2.10}$$

Since ϕ_j are eigenvalues, we have from definition of p_t, that for any $t > 0$ that

$$\phi_j(x) = e^{\lambda_j t} \int_K p_t(x,y) \phi_j(y) \, \mathrm{d}\mu(y).$$

Then, from the Cauchy-Schwartz inequality, we have for every $t > 0$

$$|\phi_j(x)| \leq e^{\lambda_j t} \left(\int_K p_t(x,y)^2 \, \mathrm{d}\mu(y) \right)^{1/2} \left(\int_K \phi_j(y)^2 \, \mathrm{d}\mu(y) \right)^{1/2} = e^{\lambda_j t} p_{2t}(x,x)^{1/2}.$$

The last equality comes from the fact that $\int_K \phi_i(y)\phi_j(y) \, \mathrm{d}\mu = \delta_{ij}$ so $\int_K \phi_j(y)^2 \, \mathrm{d}\mu(y) = 1$. That combined with (2.10) gives us $\int_K p_t(x,y)^2 \, \mathrm{d}\mu(y) = \int_K p_{2t}(x,y) \, \mathrm{d}\mu(y)$ resulting in the equality.

Now choosing $t = 1/4$ together with (2.8) that there is a constant $C > 0$ so that for every $x \in K$ we have

$$|\phi_j| \leq C e^{\lambda_j / 4}.$$

Returning to (2.10) we have for every $x, y \in K$ and $t \geq 1$ that

$$\begin{aligned} |p_t(x,y)| &\leq \sum_{j=1}^{\infty} e^{-\lambda_j t} |\phi_j(x)| |\phi_j(x)| \\ &\leq C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j / 2} \\ &= C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j / 2} e^{\lambda_j} e^{-\lambda_j} \\ &\leq C^2 e^{-\lambda_1 t} e^{\lambda_1} \sum_{j=1}^{\infty} e^{-\lambda_j / 2} \end{aligned}$$

where the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \ldots$ and $t \geq 1$ so $e^{\lambda_j (1-t)} \leq e^{\lambda_1 (1-t)}$. Thus we have the desired bound. $\square$

Proof of Proposition 2.33. We have that

$$\begin{aligned}G_s(x,y) &= \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}p_t(x,y)\ \mathrm{d}t\\ &= \frac{1}{\Gamma(s)}\int_0^1 t^{s-1}p_t(x,y)\ \mathrm{d}t + \frac{1}{\Gamma(s)}\int_1^\infty t^{s-1}p_t(x,y)\ \mathrm{d}t.\end{aligned}$$

The integral $\frac{1}{\Gamma(s)}\int_1^\infty t^{s-1}p_t(x,y)\ \mathrm{d}t$ can be uniformly bounded by a constant, using lemma 2.34. Thus we only need to bound the integral $\int_0^1 t^{s-1}p_t(x,y)\ \mathrm{d}t$. By (2.8) we have that

$$\int_0^1 t^{s-1}p_t(x,y)\ \mathrm{d}t \leq c_1\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right).$$

Since the bound of the integral relies on the value of s we will split it up into the cases of Proposition 2.33.

If $s > d_h/d_w$ then we simply have

$$\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right) \leq \int_0^1 t^{s-1-\frac{d_h}{d_w}}\ \mathrm{d}t.$$

Next, if $s < d_h/d_w$, then using a change of variable $t = ud(x,y)^{d_w}$, we find that

$$\begin{aligned}&\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right)\\ &= d(x,y)^{sd_w-d_h}\int_0^{\frac{1}{d(x,y)^{d_w}}} u^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\ &\leq d(x,y)^{sd_w-d_h}\int_0^\infty u^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u.\end{aligned}$$

Convergence of the last integral comes from the fact that $e^{-c_2(1/u)^{1/d_w}}$ converges faster than $1/u$ diverges.

Lastly for $s = d_h/d_w$, we again use the change of variable $t = ud(x,y)^{d_w}$ and setting $R = \mathrm{diam}(K)$ and noting that $0 < 1/R^{d_w} < 1/d(x,y)^{d_w}$, thus we find

$$\begin{aligned}&\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right).\\ &= \int_0^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\ &\leq \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\ &\leq \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\ \mathrm{d}u + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\ &\leq d_w\Big|\ln\left(d(x,y)\right)\Big| + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\ &\leq C\Big|\ln\left(d(x,y)\right)\Big|.\end{aligned}$$

Thus we have our desired bounds. □

2.7 Bounds of the Fractional Laplacian

In the following we will prove some bounds for the fractional Laplacian. Since the goal of this thesis is to prove convergence of fractional Gaussian fields within a specific space of test functions, we will only be proving the lemmas for that space. The space of functions is given by

Definition 2.35: We define the following space of test functions $S(K)$ as

$$S(K) = \left\{ f \in C_0(K) \colon \forall k \geq 0 \lim_{n\to\infty} n^k \left| \int_K \phi_n(y) f(y) \, \mathrm{d}\mu(y) \right| = 0 \right\}$$

with the topology defined by the family of norms

$$\|f\|_k = \left\| (-\Delta)^{-k} f \right\|_{L^2(K,\mu)}.$$

By [25, Theorem 2] and Lemma 2.29 we have that $S(K)$ is a Frèchet nuclear space. That a space is *Fréchet* means that it is nuclear and the topology is given by a countable family of seminorms. We will quickly define a nuclear space, but will not delve into the subject matter. The following definition is from [31, Chapter 50].

Definition 2.36: Let E be a locally convex Hausdorff space and $\|\cdot\|_p$ a continuous seminorm on E. Further let $\widehat{E}_p$ be a Banach space, defined to be the completion of the normed space $E \setminus \ker\|\cdot\|_p$. Then E is a *nuclear space* if to every continuous seminorm $\|\cdot\|_p$ on E there exists another continuous seminorm on E with $\|\cdot\|_q \geq \|\cdot\|_p$ such that the canonical mapping $\widehat{E}_p \to \hat{E}_q$ is *nuclear*.

A mapping $u \colon E \to F$, where E, F are Banach spaces, is *nuclear* if there is a sequence $\{x'_k\}$ in the closed unit ball of E', a sequence $\{y_k\}$ in the closed unit ball of F, and a complex sequence $\{\lambda_k\}$ with $\sum_k |\lambda_k| < \infty$ such that $u(x) = \sum_{i=1}^{\infty} \lambda_i x'_i(x) y_i$ for all $x \in X$.

Furthermore, the trace-norm $\|u\|_{\mathrm{Tr}}$ is equal to the infimum of the numbers $\sum_k |\lambda_k|$ over the set of all representations of u as such a series.

Remark: We have just defined a nuclear map on a Banach spaces, but note that the definition can be generalized to locally convex Hausdorff spaces.

The concept of a nuclear space is quite complicated, but its properties are very useful. It will allow us to prove the convergence of fractional Gaussian fields through the characteristics function and existence of fractional Gaussian fields through Bochner-Minlos theorem for nuclear spaces, which will be state later in Theorem 3.5. But first we will prove some lemmas regarding the fractional Laplacian and the fractional Riesz kernel.

Lemma 2.37: Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s} f = \int_K G_s(x, y) f(y) \, \mathrm{d}\mu(y).$$

Proof. First, we will make sure the integral on the right side actually converges. This holds because $S(K) \subset C(K)$ whereby f will be bounded on K. Thus we just need for every $x \in K$, that $\int_K G_s(x, y) \, \mathrm{d}\mu(y)$ is convergent.[1] For $s \in (d_h/d_w, \infty)$ the convergence follows from proposition 2.33. Looking at $s \in (0, d_h/d_w]$, we will use a dyadic annuli decomposition. Meaning

[1]In Proposition 2.33, (i) and (ii) the bounds are for $x \neq y$, but they are null-sets with respect to μ

we will split K up into infinitely many annuli on the form $B(x, R2^{-j}) \setminus B(x, RS^{-j-1})$, noting that

$$K \subseteq \bigcup_{j \geq 0} B(x, R2^{-j}) \setminus B(x, RS^{-j-1})$$

where $R = \mathrm{diam}(K)$. Then we have for all $y \in B(x, R2^{-j}) \setminus B(x, RS^{-j-1})$ that

$$R2^{-j-1} \leq d(x,y) \leq R2^{-j-1} \Rightarrow 2^{-j-1} \leq R^{-1} \cdot d(x,y) \leq 2^{-j-1}.$$

We will show that the integral is finite for both $s = d_h/d_w$ and $s < d_h/d_w$, but this requires we bound $\Big|\ln\left(d(x,y)\right)\Big|$. So we see that for all $\alpha > 0$

$$x^{\alpha}\big|\ln(x)\big| \to 0, \quad \text{as } x \to 0$$

which means there exists a constant $k > 0$ such we can choose some interval $(0, u_k]$ so $\ln(x) \leq \frac{k}{x^{\alpha}}$ for all $x \in (0, u_K]$. Now choose k such $u_{k'} \leq R$, in which case for $s = d_h/d_w$ we have

$$\begin{aligned}
\int_K G_s(x,y)\ \mathrm{d}\mu(y) &\leq \int_K C_1 \cdot \Big|\ln\left(d(x,y)\right)\Big|\ \mathrm{d}\mu(y) \\
&\leq \sum_{j=0}^{\infty} \int_{B(x,R2^{-j}) \setminus B(x,RS^{-j-1})} C_1 \cdot \Big|\ln\left(d(x,y)\right)\Big|\ \mathrm{d}\mu(y) \\
&\leq \sum_{j=0}^{\infty} \int_{B(x,R2^{-j}) \setminus B(x,RS^{-j-1})} \frac{C_1 \cdot k'}{d(x,y)^{\gamma}}\ \mathrm{d}\mu
\end{aligned}$$

where $\gamma < d_h$. We can likewise bound $G_s(x,y)$ by

$$\begin{aligned}
\int_K G_s(x,y)\ \mathrm{d}\mu(y) &\leq \int_K \frac{C_2}{d(x,y)^{\gamma}}\ \mathrm{d}\mu(y) \\
&\leq \sum_{j=0}^{\infty} \int_{B(x,R2^{-j}) \setminus B(x,RS^{-j-1})} \frac{C}{d(x,y)^{\gamma}}\ \mathrm{d}\mu(y)
\end{aligned}$$

where $C = \max\left\{C_1 \cdot k', C_2\right\}$. Thus for $s \in (0, d_h/d_w]$ we have

$$\begin{aligned}
\int_K G_s(x,y)\ \mathrm{d}\mu(y) &\leq \sum_{j=0}^{\infty} \int_{B(x,R2^{-j}) \setminus B(x,RS^{-j-1})} \frac{C}{d(x,y)^{\gamma}}\ \mathrm{d}\mu(y) \\
&\leq R \cdot C \sum_{j=0}^{\infty} 2^{j\gamma} \mu\left(B(x,R2^{-j}) \setminus B(x,RS^{-j-1})\right) \\
&\leq R \cdot C \sum_{j=0}^{\infty} 2^{j\gamma} \mu\left(B(x,R2^{-j})\right) \\
&\leq R^{d_h+1} \cdot C \cdot c_2 \sum_{j=0}^{\infty} 2^{j(\gamma - d_h)} < \infty
\end{aligned}$$

Where we use proposition 1.14 and that $\int_K G_s(x,y)\ \mathrm{d}\mu(y)$ is convergent for every $x \in K$.

Moving along in proving the lemma, we note that if we use the substitution $t = \lambda_j u$ we get

$$\frac{1}{\Gamma(s)} \int_0^{\infty} t^{s-1} e^{-\lambda_j t}\ \mathrm{d}t = \frac{1}{\Gamma(s) \cdot \lambda_j^s} \int_0^{\infty} u^{s-1} e^{-u}\ \mathrm{d}u = \frac{1}{\Gamma(s)\lambda_j^s}\Gamma(s) = \lambda_j^{-s}.$$

Now using Fubini's theorem, Definition 2.32 and theorem 2.28 we get

$$\begin{aligned}
\int_K G_s(x,y)f(y)\ \mathrm{d}\mu(y) &= \int_K \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}p_t(x,y)\ \mathrm{d}t f(y)\ \mathrm{d}\mu(y) \\
&= \int_K \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\sum_{j=1}^\infty e^{-\lambda_j t}\phi_j(x)\phi_j(y)\ \mathrm{d}t f(y)\ \mathrm{d}\mu(y) \\
&= \sum_{j=1}^\infty \phi_j(x)\int_K \phi_j(y)f(y)\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}e^{-\lambda_j t}\ \mathrm{d}t\ \mathrm{d}\mu(y) \\
&= \sum_{j=1}^\infty \frac{1}{\lambda_j^s}\phi_j(x)\int_K \phi_j(y)f(y)\ \mathrm{d}\mu(y) \\
&= (-\Delta)^{-s}f
\end{aligned}$$

as desired. □

An immediate corollary follows:

Corollary 2.38: Let $(-\Delta)^{-s}$ be defined as in Definition 2.30, then the following holds

(i) Let $s > 0$, for $f, g \in S(K)$

$$\int_K (-\Delta)^{-s}f(-\Delta)^{-s}g\ \mathrm{d}\mu = \int_K\int_K f(x)g(y)G_{2s}(x,y)\ \mathrm{d}\mu(x)\ \mathrm{d}\mu(y)$$

(ii) Let $s > 0$ the for $f \in S(K)$

$$(-\Delta)^{-s}f = \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}P_t f(x)\ \mathrm{d}t$$

Proof. Beginning with (i). It follows from the definition of $(-\Delta)^{-s}$, Proposition 2.31 and Lemma 2.37, we find that

$$\begin{aligned}
\int_K (-\Delta)^{-s}f(-\Delta)^{-s}g\ \mathrm{d}\mu &= \left\langle (-\Delta)^{-s}f, (-\Delta)^{-s}g\right\rangle \\
&= \left\langle f, (-\Delta)^{-2s}g\right\rangle \\
&= \int_K f(-\Delta)^{-2s}g\ \mathrm{d}\mu \\
&= \int_K\int_K f(x)g(y)G_{2s}(x,y)\ \mathrm{d}\mu(x)\ \mathrm{d}\mu(y)
\end{aligned}$$

Moving on to (ii). Again from Lemma 2.37 and theorem 2.28 we have

$$\begin{aligned}
(-\Delta)^{-s}f &= \sum_{j=1}^\infty (\lambda_j)^{-s}\phi_j\int_K \phi_j(y)f(y)\ \mathrm{d}\mu(y) \\
&= \sum_{j=1}^\infty \left(\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}e^{-\lambda_j t}\ \mathrm{d}t\right)\phi_j\int_K \phi_j(y)f(y)\ \mathrm{d}\mu(y) \\
&= \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\int_K\sum_{j=1}^\infty e^{-\lambda_j t}\phi_j\phi_j(y)f(y)\ \mathrm{d}\mu(y)\ \mathrm{d}t \\
&= \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}\int_K p_t(x,y)f(y)\ \mathrm{d}\mu(y)\ \mathrm{d}t \\
&= \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}P_t f(x)\ \mathrm{d}t
\end{aligned}$$

□

Chapter 3

Fractional Gaussian Fields

In this chapter we will be constructing both the fractional Gaussian fields and the discrete fractional Gaussain fields, that we will later see converge in distribution. The construction of the fractional Gaussian fields very much depend on the Riesz kernel and the Dirichlet Laplacian, with the discrete fractional Gaussian field depending on the discrete fractional Laplacian and discrete Riesz kernel.

We will also give an example of an discrete fractional Gaussian field, which we will use in some simulations.

3.1 Discrete Fractional Gaussian Fields

Definition 3.1: Let $s \geq 0$, then a *discrete fractional Gaussian field* X_s^m with parameter s on V_m is a Gaussian vector indexed by V_m with mean zero and covariance matrix $\mathbf{\Sigma} = \left(G_{2s}^m(x,y)\right)_{x,y \in V_m}$.

Notation: For any $f \in \ell(V_m)$ we will use the notation

$$X_s^m(f) = \frac{1}{a_m} \sum_{p \in V_m} f(p) X_s^m(p).$$

We then note that the discrete fractional Gaussian field has the following property:

Proposition 3.2: For $f, g \in \ell(V_m)$ we have that

$$\mathbb{E}\left[X_s^m(f) X_s^m(g)\right] = \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p)$$

Proof. Using Proposition 2.22 we find

$$\begin{aligned}
\mathbb{E}\left[X_s^m(f) X_s^m(g)\right] &= \frac{1}{2a_m} \sum_{p,q \in V_m} f(p) g(q) \mathbb{E}\left[X_s^m(p) X_s^m(q)\right] \\
&= \frac{1}{2a_m} \sum_{p,q \in V_m} f(p) g(q) G_{2s}^m(p,q) \\
&= \frac{1}{a_m} \sum_{p \in V_m} f(p) \left(\frac{1}{a_m} \sum_{q \in V_m} g(q) G_{2s}^m(p,q) \right) \\
&= \frac{1}{a_m} \sum_{p \in V_m} f(p) (-\Delta_m)^{-2s} g(p) \\
&= \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p)
\end{aligned}$$

□

Example 3.3: Here we will give an example of a discrete fractional Gaussian field. Let $(W_i)_{1\le i\le N_m}$ be a sequence of i.i.d Gaussian random variables with mean 0 and variance 1, i.e. white noise. Then define

$$X_s^m(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \, \phi_i^m(x) W_i, \quad x \in V_m.$$

Initially, we check the mean, so

$$\mathbb{E}\left[X_s^m\right] = \begin{pmatrix} \mathbb{E}\left[X_s^m(x_1)\right] \\ \vdots \\ \mathbb{E}\left[X_s^m(x_{N_m})\right] \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

since

$$\mathbb{E}\left[X_s^m(x_i)\right] = \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \, \phi_i^m(x) W_i\right] = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \, \phi_i^m(x) \mathbb{E}\left[W_i\right] = 0.$$

Moving on to the covariance matrix, we have that

$$\mathbf{\Sigma} = \mathbb{E}\left(\left(X_s^m - \mathbb{E}\left[X_s^m\right]\right)\left((X_s^m)^T - \mathbb{E}\left[X_s^m\right]\right)\right) =$$

$$\mathbb{E}\left[X_s^m \cdot (X_s^m)^T\right] = \mathbb{E}\left[\begin{pmatrix} X_s^m(x_1)X_s^m(x_1) & \cdots & X_s^m(x_1)X_s^m(x_{N_m}) \\ \vdots & \ddots & \vdots \\ X_s^m(x_{N_m})X_s^m(x_1) & \cdots & X_s^m(x_{N_m})X_s^m(x_{N_m}) \end{pmatrix}\right].$$

Looking at a single element $\mathbf{\Sigma}_{p,q}$, and using that W_i are independent, we compute

$$\begin{aligned}
\mathbb{E}\left[X_s^m(x_p)X_s^m(x_q)\right] &= \mathbb{E}\left[\left(\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \, \phi_i^m(x_p) W_i\right)\left(\sum_{j=1}^{N_m} \left(\lambda_j^m\right)^{-s} \, \phi_j^m(x_q) W_j\right)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \, \phi_i^m(x_p)\phi_i^m(x_q) W_i^2\right] + \mathbb{E}\left[\sum_{i,j=1, i\neq j}^{N_m} \left(\lambda_i^m \lambda_j^m\right)^{-s} \, \phi_i^m(x_p)\phi_j^m(x_q) W_i W_j\right] \\
&= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \, \phi_i^m(x_p)\phi_i^m(x_q) \mathbb{E}\left[W_i^2\right] + \sum_{i,j=1, i\neq j}^{N_m} \left(\lambda_i^m \lambda_j^m\right)^{-s} \, \phi_i^m(x_p)\phi_j^m(x_q) \mathbb{E}\left[W_i\right]\mathbb{E}\left[W_j\right] \\
&= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \, \phi_i^m(x_p)\phi_i^m(x_q) \\
&= G_{2s}^m(x_p, x_q).
\end{aligned}$$

Thus X_s^m is a discrete fractional Gaussian field. ◊

3.2 Fractional Gaussian Fields

Definition 3.4: Let $s \ge 0$. A *fractional Gaussian field* X_s with parameter s on K is a centered Gaussian field, $\{X_s(f) : f \in S(K)\}$, such that for any $f, g \in S(K)$

$$\mathbb{E}\left[X_s(f)X_s(g)\right] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \; \mathrm{d}\mu.$$

We now move on to proving that the fractional Gaussian fields exist. First we will state the Bochner-Minlos theorem for nuclear spaces, since it is needed for the proof of existence. It will be given without proof but one can be found in [19, Theorem 2.3.1].

Theorem 3.5 (Bochner-Minlos)**:** Assume $\mathcal{X}$ is a nuclear space. Any continuous positive-definite linear functional Λ on $\mathcal{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathcal{X}'$.

In the next lemma we will prove that a functional, which we will need, is positive semidefinite.

Lemma 3.6: The functional $\Phi : \mathcal{X} \to \mathbb{R}$ defined as $\Phi(v) = \exp\left(-\frac{1}{2}\langle v, v\rangle\right)$ is positive definite.

Proof. Let $v_1, \dots v_n$ be elements of $\mathcal{X}$ and choose an orthonormal basis $e_1, \dots e_m$ of span $\{v_1, \dots, v_n\}$. Further let $Z = (Z_1, \dots, Z_m)$ be a vector of independent standard normal random variables, and note that for all $u \in \mathbb{R}^m$ we have that

$$\Phi\left(\sum_{j=1}^{m} u_j e_j\right) = \exp\left(-\frac{1}{2}\sum_{i=1}^{m} u_i^2\right) = \mathbb{E}\left[e^{\mathrm{i}u\cdot Z}\right].$$

This leads us to for $z \in \mathbb{C}^n$

$$\begin{aligned}\sum_{j,k=1}^{n} z_j\overline{z_k}\Phi(v_j - v_k) &= \sum_{j,k=1}^{n} z_j\overline{z_k}\mathbb{E}\left[e^{\mathrm{i}(v_j-v_k)\cdot Z}\right] \\ &= \mathbb{E}\left[\left(\sum_{j=1}^{n} z_j e^{iv_j\cdot Z}\right)\left(\sum_{k=1}^{n}\overline{z_k e^{iv_k\cdot Z}}\right)\right] = \mathbb{E}\left[\left|\sum_{j=1} z_j e^{iv_j\cdot Z}\right|^2\right] \geq 0.\end{aligned}$$

□

Theorem 3.7: Let $s \geq 0$. There exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E}\left[X_s(f)X_s(g)\right] = \int_K (-\Delta)^{-s} f(-\Delta)^{-s} g \; \mathrm{d}\mu.$$

Proof. The space $S(K)$ is a nuclear space, why it is enough to use Theorem 3.5 (Bochner-Minlos) on the functional

$$\Lambda : f \to \exp\left(-\frac{1}{2}\|f\|_k^2\right).$$

Since if Theorem 3.5 holds, then it gives us a unique probability measure ν such that

$$\exp\left(-\frac{1}{2}\|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x,f\rangle} \; \mathrm{d}\nu(x).$$

Notice that the right side is a characteristic function for some random variable. If we choose $f = t \cdot g$ for some $g \in S(K)$ and $t \in \mathbb{R}$, note that $t \cdot g \in S(K)$, then the left side of the equation becomes the characteristic function for the normal distribution with mean zero and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now we prove that Λ satisfies the requirements of Theorem 3.5. Obviously $\Lambda(0) = 1$, and from lemma 3.6 Λ is positive definite. Lastly we need to prove it is continuous at 0.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C > 0$ so $\left\|(-\Delta)^{-s} f\right\|_{L^2(K,\mu)} \leq C\|f\|_{L^2(K,\mu)}$. Now let $\varepsilon > 0$ be given and choose

$$\delta = \begin{cases} \left(\frac{-2\cdot\ln(1-\varepsilon)}{C}\right)^{\frac{1}{2}}, & \text{when } \varepsilon \in (0,1) \\ 1, & \text{when } \varepsilon \geq 1. \end{cases}$$

If $\varepsilon \geq 1$ notice that

$$\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| \leq 1 \leq \varepsilon, \quad \forall f \in S(K)$$

in which case we can choose δ to be any real number larger than 0. Moving on to $\varepsilon \in (0,1)$ we get

$$\begin{aligned}
\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| &\leq \left|\exp\left(-\frac{1}{2}C\|f\|_{L^2(K,\mu)}\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K |f|^2 \ \mathrm{d}\mu\right) - 1\right| \\
&\leq \left|\exp\left(-\frac{1}{2}C\int_K \delta^2 \ \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K \frac{-2\cdot\ln(1-\varepsilon)}{C} \ \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(\ln(1-\varepsilon)\right) - 1\right| \\
&= \varepsilon
\end{aligned}$$

using that μ is a probability measure in the third to last equality. In which case we have continuity in 0 as desired. □

In the following we want to prove that a fractional Gaussian field has a L^2 density if and only if $s > \frac{d_h}{2d_w}$, but for this we need white noise.

Definition 3.8: Let $(X, \mathcal{E}, \mu)$ be σ-finite measure space. A *white noise based on* ν is a random set function W on the sets $A \in \mathcal{E}$ with $\nu(A) < \infty$ such that

(i) $W(A)$ is a $N(0, \mu(A))$ random variable.

(ii) If $A \cap B = \emptyset$ then $W(A)$ and $W(B)$ are independent and $W(A \cap B) = W(A) + W(B)$.

We will now show that such a process exists on $(K, \mathcal{K}, \mu)$ where $\mathcal{K}$ is the Borel σ-algebra on K.

We can think of white noise as a Gaussian process indexed by the set

$$\mathcal{A} = \{W(A) : A \in \mathcal{K}, \mu(A) < \infty \ \}.$$

From Definition 3.8, (i) and (ii) it is a mean-zero Gaussian process with covariance matrix

$$\mathbf{\Sigma} = \left(\mathbb{E}\left[W(A)W(B)\right]\right)_{W(A),W(B)\in\mathcal{A}} = \left(\mu(A\cap B)\right)_{W(A),W(B)\in\mathcal{A}}.$$

Now let $A_1, \ldots, A_n \in \mathcal{K}$ and $a_1, \ldots, a_n$ be real numbers. Then

$$\sum_{i,j} a_i a_j \mathbf{\Sigma}_{i,j} = \sum_{i,j} a_i a_j \int_K \mathbb{1}_{A_i} \cdot \mathbb{1}_{A_j} \ \mathrm{d}\mu = \int_K \left(\sum_i a_i \mathbb{1}_{A_i}\right)^2 \ \mathrm{d}\mu \geq 0$$

Thus $\mathbf{\Sigma}$ is positive definite, so there exists a probability space $(\Omega, \mathcal{F}, p)$ and a mean-zero Gaussian process $\{W(A)\}$ on $(\Omega, \mathcal{F}, p)$ such that W satisfies Definition 3.8 (i) and (ii).

This means that for $f \in L^2(K, \mathcal{K}, \mu)$ the integral with respect to W, $W(f) = \int_K f \, \mathrm{d}W$ is a well defined Gaussian random variable. Moreover, the Gaussian measure W gives rise to an isonormal Gaussian family $\left\{W(f), f \in L^2(K, \mathcal{K}, \mu)\right\}$ with the covariance function

$$\mathbb{E}\left[W(f)W(g)\right] = \int_K fg \, \mathrm{d}\mu.$$

In the following we want to define a fractional Brownian field, and we want to do it as the integral of the fractional Riesz kernel defined in definition 2.32, thus we need $G_s(x, \cdot) \in L^2(K, \mu)$.

Let $s > \frac{d_h}{2d_w}$. We observe from proposition 2.33 that $G_s(x, \cdot) \in L^2(K, \mu)$. Moreover we have that

$$G_s(x,y) = \frac{1}{\Gamma(s)} \int_K t^{s-1} p_t(x,y) \, \mathrm{d}t = \frac{1}{\Gamma(s)} \int_K t^{s-1} \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x)\phi_j(y) \, \mathrm{d}t = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{-s}} \phi_j(x)\phi_j(y). \tag{3.1}$$

recalling that 2.9 and $\frac{1}{\Gamma(s)} \int_K t^{s-1} e^{-\lambda_j t} \, \mathrm{d}t = \lambda_j^{-s}$. Furthermore note that by Lemma 2.29 we have (3.1) converges, since $s > \frac{d_h}{2d_w}$.

Definition 3.9: Let $s > \frac{d_h}{2d_w}$. The fractional Brownian field with parameter s is defined as

$$\widetilde{X}_s(x) = \int_K G_s(x,y) \, \mathrm{d}W(y)$$

Remark 3.10: Since $s > \frac{d_h}{2d_w}$ then by (3.1) we can write

$$\widetilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s} \phi_i(x) W_i$$

where $W_i = W(\phi_i)$ is an i.i.d sequence of Gaussian random variables with mean 0 and variance 1.

We will now show that one can construct a fractional Gaussian field from the fractional Brownian field.

Proposition 3.11: Let $s > \frac{d_h}{2d_w}$. Then the Gaussian random field defined by

$$X_s(f) = \int_K f(x)\widetilde{X}_s(x) \, \mathrm{d}\mu(x), \quad f \in S(K)$$

has the law of a fractional Gaussian field with parameter s. Moreover, if there exists a Gaussian field $\left(\widetilde{X}_s(x)\right)_{x \in K}$ on K with

$$\mathbb{E}\left[\int_K \widetilde{Y}_S(x)^2 \, \mathrm{d}\mu(x)\right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x)\widetilde{Y}_s(x) \, \mathrm{d}\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s, then $s > \frac{d_h}{2d_w}$.

Proof. Assume $s > \frac{d_h}{2d_w}$. Then from Fubini's theorem we have for every $f \in S(K)$ a.s. that

$$\int_K f(x)\widetilde{X}_s(x) \, \mathrm{d}\mu(x) = \int_K f(x) \int_K G_s(x,y) \, \mathrm{d}W(y) \, \mathrm{d}\mu(x) = \int_K (-\Delta)^{-s} f(y) \, \mathrm{d}W(y)$$

in which case by definition of W we have that $X_s(f) = \int_K f(x)\widetilde{X}_s(x)\ \mathrm{d}\mu(x)$ is a Gaussian random variable with mean 0 and variance

$$\int_K \left|(-\Delta)^{-s} f(x)\right|^2\ \mathrm{d}\mu(x)$$

and thus a fractional Gaussian field with parameter s.

Now assume that there is a Gaussian field $\left(\widetilde{X}_s(x)\right)_{x\in K}$ on K with

$$\mathbb{E}\left[\int_K \widetilde{Y}_S(x)^2\ \mathrm{d}\mu(x)\right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x)\widetilde{Y}_s(x)\ \mathrm{d}\mu(x), \quad f \in S(K) \tag{3.2}$$

has the law of fractional Gaussian field with parameter s. Then by spectral decomposition and equation (3.2) we have

$$\widetilde{Y}_s(x) = \sum_{i=1}^{\infty} \left\langle \widetilde{Y}_s, \phi_i \right\rangle \phi_i(x) = \sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x).$$

Note that $(Y(\phi_i))_{i\geq 1}$ is a sequence of independent Gaussian random variables with mean 0 and variance $(\lambda_i^{-2s})_{i\geq 1}$ since by definition 3.4 and proposition 2.31 we have that

$$\mathbb{E}\left[Y_s(\phi_i)\right] = \mathbb{E}\left[\int_K \phi_i \widetilde{Y}_s\ \mathrm{d}\mu\right] = \mathbb{E}\left[\int_K \phi_i(x)\sum_{j=1}^{\infty} \lambda_j^{-s}\phi_j(x)W_i\ \mathrm{d}\mu(x)\right] = \mathbb{E}\left[W_i\right] = 0$$

and

$$\mathbb{E}\left[Y_s(\phi_i)Y_s(\phi_j)\right] = \int_K (-\Delta)^{-s}\phi_i(-\Delta)^{-s}\phi_j\ \mathrm{d}\mu = \frac{1}{(\lambda_j\lambda_i)^s}\int_K \phi_i\phi_j\ \mathrm{d}\mu = \frac{1}{(\lambda_j\lambda_i)^s}\delta_{ij}.$$

Since $\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] < \infty$ we find that

$$\begin{aligned}
\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] &= \mathbb{E}\left[\int_K \left(\sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x)\right)^2\ \mathrm{d}\mu(x)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} Y_s(\phi_i)^2 \int_K \phi_i(x)\phi_i(x)\ \mathrm{d}\mu(x)\right] \\
&= \sum_{i=1}^{\infty} \mathbb{E}\left[Y_s(\phi_i)^2\right] \\
&= \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{-2s}} < \infty
\end{aligned}$$

where we use that $\int_K \left(\sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x)\right)^2\ \mathrm{d}\mu(x) = \int_K \sum_{i=1}^{\infty} Y_s(\phi_i)^2\phi_i(x)^2\ \mathrm{d}\mu(x)$ which stems from the fact that $\int_K \phi_i\phi_j\ \mathrm{d}\mu = \delta_{ij}$ and $Y_s(\phi_i)$ is a constant with respect to $\mu(x)$.

We now have that

$$\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] = \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{-2s}} < \infty$$

in which case by Lemma 2.29 we have that $s > \frac{d_h}{2d_w}$. □

Chapter 4

Convergence in distribution of fractional Gaussian Fields

Now that we have presented and proved the existence of fractional Gaussian fields X_s on K and discrete fractional Gaussian fields X_s^m on V_m, our next task is proving that they converge. Therefore, in this section our main goal is to show that X_s^m converges in distribution to X_s.

4.1 Preliminary Lemmas

In this section we will prove a collection of lemmas that will be needed.

Lemma 4.1: Let $\{P_t\}_{t\geq 0}$ be the semigroup generated by the Dirichlet Laplacian $(-\Delta)^{-s}$ and $p_t(x,y)$ the heat kernel admitted by $\{P_t\}_{t\geq 0}$. If ϕ_i is an orthonormal eigenfunction of $(-\Delta)^{-s}$ then

$$P_t\phi_i(x) = e^{-\lambda_i t}\phi_i(x).$$

Proof. The lemma follows immediately from the calculations:

$$\begin{aligned} P_t\phi_i(x) = \int_K p_t(x,y)\phi_i(y)\ \mathrm{d}\mu(y) &= \int_K \left(\sum_{j=1}^{\infty} e^{-\lambda_j}\phi_j(x)\phi_j(y)\right)\phi_i(y)\ \mathrm{d}\mu(y) \\ &= \int_K e^{-\lambda_i t}\phi_i(x)\ \mathrm{d}\mu(y) = e^{-\lambda_i t}\phi_i(x) \end{aligned}$$

□

Lemma 4.2: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m\to\infty} \frac{1}{a_m}\sum_{p\in V_m} f_m(p)P_t^m f_m(p) = \int_K f(x)P_t f(x)\ \mathrm{d}\mu(x)$$

where $f_m = f\mid_{V_m}$.

Proof. In this proof we wish to use [21, Theorem 2.1] so we will check that Δ fulfils the criteria. First there is a bounded linear operator such that for all $f \in L^2(K,\mu)$ we have

$$\lim_{n\to\infty}\|P_n f\| = \|P_n\|.$$

This holds in our case, since we have $P_n f = f\mid_{V_n} = f_n$.

Secondly we note that Δ is the infinitesimal generator of a uniformly continuous semigroup by Theorem 2.24.

Next we have that the sequence of $\{\Delta_m\}_{m\geq 0}$ indeed coincides with the extended limit defined, in [21, p. 355], this follows from Theorem 2.16 and Section 2.3.

Finally the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 by [3, Lemma 1.1.1].
Thus by [21, Theorem 2.1] we have that for all $t \geq 0$

$$\lim_{m\to\infty} \sup_{p\in V_m} \left| P_t^m f_m(p) - \left(P_t f(p)\right)_m \right| = 0. \tag{4.1}$$

If we now write

$$\frac{1}{a_m} \sum_{p\in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \, \mathrm{d}\mu_m$$

then

$$\int_{V_m} f_m P_t^m f_m \, \mathrm{d}\mu_m = \int_{V_m} f_m \left(P_t^m f_m - (P_t f)_m\right) \, \mathrm{d}\mu_m + \int_{V_m} f_m (P_t f)_m \, \mathrm{d}\mu_m.$$

The first integral converges to 0 as $m \to \infty$ because of (4.1). Further, by corollary 1.16 we have $\int_{V_m} f_m (P_t f)_m \, \mathrm{d}\mu_m \to \int_K f P_t f \, \mathrm{d}\mu$ as $m \to \infty$. □

Lemma 4.3: For all $f \in S(K)$ and $s \geq 0$ the following holds

$$\frac{1}{a_m} \sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) \to \int_K f(x)(-\Delta)^{-2s} f(x) \, \mathrm{d}\mu(x), \quad \text{as } m \to \infty.$$

Proof. For $s = 0$ we find that

$$\begin{aligned}
(-\Delta_m)^{-0} f &= \sum_{i=1}^{N_m} \frac{1}{(\lambda_i^m)^0} \phi_i^m(x) \frac{1}{a_m} \sum_{y\in V_m} \phi_i^m(y) f(y) \\
&= \sum_{i=1}^{N_m} \phi_i^m(x) \int_{V_m} \phi_i^m(y) f(y) \, \mathrm{d}\mu_m \\
&= f_m.
\end{aligned}$$

Likewise by Definition 2.30 we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, \mathrm{d}\mu = f$$

and the lemma for $s = 0$ follows from Corollary 1.16.

Now assume $s > 0$, then we first note that

$$\begin{aligned}
(-\Delta_m)^s f_m &= \frac{1}{a_m} \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x) \sum_{y\in V_m} \phi_j(y) f_m(y) \\
&= \frac{1}{a_m} \sum_{j=1}^{N_m} \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, \mathrm{d}t \; \phi_j^m(x) \sum_{y\in V_m} \phi_j(y) f_m(y) \\
&= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \frac{1}{a_m} \sum_{y\in V_m} \sum_{j=1}^{N_m} e^{-\lambda_j t} \phi_j^m(x) \phi_j(y) f_m(y).
\end{aligned}$$

Now by Fubini's theorem and Corollary 2.25 we have that

$$\frac{1}{a_m}\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)}\int_0^\infty t^{2s-1}\frac{1}{a_m}\sum_{p\in V_m} f_m(p)P_t^m f_m(p).$$

By Lemma 4.1 we find that

$$\begin{aligned}\frac{1}{a_m}\sum_{p\in V_m} f_m(p)P_t^m f_m(p) &= \frac{1}{a_m}\sum_{p\in V_m} f_m(p)P_t^m\sum_{j=1}^\infty \langle f_m,\phi_i\rangle\,\phi_i\\ &= \frac{1}{a_m}\sum_{p\in V_m} f_m(p)\sum_{j=1}^\infty \langle f_m,\phi_j\rangle\,P_t^m\phi_j\\ &= \frac{1}{a_m}\sum_{p\in V_m} f_m(p)\sum_{j=1}^\infty \langle f_m,\phi_j\rangle\,e^{-\lambda_j^m t}\phi_j\\ &\le e^{-\lambda_1^m t}\frac{1}{a_m}\sum_{p\in V_m} f_m(p)^2\end{aligned}$$

where we can guarantee the inequality for all j since $0<\lambda_1<\lambda_2<\dots$.

Note that $\sup_m \frac{1}{a_m}\sum_{p\in V_m} f_m(p)^2<\infty$ since $f\in S(K)\subset C(K)$ whereby f is bounded and $\#V_m<\infty$ for all m. We can now by dominated convergence, Fubini's theorem and Lemma 4.2 find that

$$\begin{aligned}\lim_{m\to\infty}\frac{1}{a_m}\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)}\int_0^\infty t^{2s-1}t^{2s-1}\lim_{m\to\infty}\frac{1}{a_m}\sum_{p\in V_m} f_m(p)P_t^m f_m(p)\ \mathrm{d}t\\ &= \frac{1}{\Gamma(2s)}\int_0^\infty t^{2s-1}\int_K f(x)P_t f(x)\ \mathrm{d}\mu(x)\ \mathrm{d}t\\ &= \frac{1}{\Gamma(2s)}\int_K f(x)\int_0^\infty t^{2s-1}P_t f(x)\ \mathrm{d}t\ \mathrm{d}\mu(x)\\ &= \int_K f(x)(-\Delta)^{-2s}f(x)\ \mathrm{d}\mu(x)\end{aligned}$$

as desired. □

4.2 Main Theorem

With everything in mind we can now prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields.

Main Theorem 4.1: Let $s\ge 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m\to\infty$

Proof. Since $S(K)$ is a nuclear space, by [9, Corollary 2.4] it is enough to prove the convergence of the characteristic functional, e.i that for all $f\in S(K)$

$$\mathbb{E}\left[\exp\left(itX_s^m(f_m)\right)\right]\to\mathbb{E}\left[\exp\left(itX_s(f)\right)\right],\qquad \text{as } m\to\infty$$

where $f_m = f\,|_{V_m}$. It follows from Proposition 3.2 that

$$\mathbb{E}\left[\exp\left(itX_s^m(f_m)\right)\right]=\exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s^m(f_m))^2\right]\right)=\exp\left(-\frac{1}{2a_m}t^2\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s}f_m(p)\right).$$

Likewise, via Definition 3.4 we find

$$\mathbb{E}\left[\exp(itX_s(f))\right] = \exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s(f))^2\right]\right) = \exp\left(-\frac{1}{2}t^2\int_K f(-\Delta)^{-2s}f \ \mathrm{d}\mu\right)$$

where we notice bycorollary 2.38 that

$$\begin{aligned}\mathbb{E}\left[(X_s(f))^2\right] &= \int_K (-\Delta)^{-s}f(-\Delta)^{-s}f \ \mathrm{d}\mu \\ &= \int_K\int_K f(p)f(q)G_{2s}(p,q) \ \mathrm{d}\mu(p) \ \mathrm{d}\mu(q) \\ &= \int_K f(-\Delta)^{-2s}f \ \mathrm{d}\mu.\end{aligned}$$

The proof now follows from lemma 4.3. □

Chapter 5

Simulations

As a part of the thesis we have created and simulated a discrete fractional Gaussian field on the Vicsek set V_m. The precise field we have constructed was the one given in Example 3.3. These simulations have had a varying rate of success. We have been able to simulate $X_s^m(x) = \sum \lambda_j \phi_j$ for a reasonable set of s-values, but we have encountered a limiting factor in the fact that we are not able to simulate X_s^m on the Vicsek set for $m \geq 7$. The reason for this limitation lies in Remark 2.2.

As stated in the remark for any function $f\colon V_m \to \mathbb{R}$ we can define a matrix L_m such that

$$L_m F = \begin{pmatrix} \Delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

It is precisely from this matrix L_m we compute the eigenvalues and eigenfunctions of $-\Delta_m$. Note that the amount of points in V_m is $5^m \cdot 4 + 1$ so as m grows this number explodes, see Table 5.1), and there is no clever way to construct the matrix L_m. Thus the method of constructing the matrix L_m was simply by constructing the Vicsek set V_m and manually checking each point for neighbours.

m	$\#V_m$
1	21
2	101
3	501
4	2501
5	12501
6	62501
7	312501
10	39062501
20	$3.81 \cdot 10^{14}$

Table 5.1: Points in V_m.

As we see from Table 5.1, the values explode in size as m grows. The process to check each point is therefore slow. Furthermore the matrix L_m that we construct becomes very large, very fast. Since it needs to hold every permutation of points, L_m becomes a $(5^m \cdot 4 + 1) \times (5^m \cdot 4 + 1)$ matrix. It does however have the property of a sparse matrix, since there is at most five entries in every column, but for $m = 9$ the matrix still fills well over 1 terrabyte in memory.

Having explained the limitations of the simulation, we will quickly delve into the code that generates the simulations. The code will be explained mathematically for clarity, but the full code can be found at:
`https://gitfront.io/r/Cataclisem/CxGoTRPqvjoM/GaussianFieldsThesis/`.

5.1 Construction of the Vicsek set for simulation

The construction is done by first generating the first "plus" $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ where $p_1 = (0,0)$, $p_2 = (\xi, 0)$, $p_3 = (0, -\xi)$, $p_4 = (-\xi, 0)$ and $p_5 = (0, \xi)$ for some $\xi > 0$. Then going from V_0 to V_1 we find the endpoints, the tips of the plus, of V_0 and then just add V_0 to it. Formally written we have:

Let $V_m^{(0)}$ be the endpoints after iteration m and V_m be the points of the Vicsek set after iteration m, then we define a recursive function so

$$V_m = \left\{ y + 2 \cdot x : x \in V_{m-1}, y \in V_{m-1}^{(0)} \right\}, \quad \text{and} \quad V_m^{(0)} = \left\{ 3y : y \in V_{m-1}^{(0)} \right\}.$$

Recall from Definition 3.1 that a discrete fractional Gaussian field is a Gaussian vector indexed by V_m, since the discrete Laplacian depends on the points being in the same places. Thus if we were to compare two different discrete fields we need to make sure the points are in the same order in the list in Python. Luckily we can use the `set` function in Python. Since the `set` function hashes each point, it will put the point in a random place in the set, but it will put the point in the same spot every time, given that ξ and m is the same each time.

The next step is finding all the neighbours of each point. As stated before, there is no good way to find the neighbours thus this step just iterates over every point. We are therefore going to abuse the symmetry and that we have constructed the Vicsek set outwards. This means that all points in V_m, where $x \underset{m}{\sim} y$, are exactly $|x - y| = \xi$ from each other. Let $V_{m,p}$ denote the neighbours of a point p. We thus have

$$V_{m,p} = \{ x + v \in V_m : x \in V_m, v \in V_0 \}.$$

So we actually just add ξ in every direction and see if the point is contained in V_m and finding the neighbours. In Python, we collect each point along with it neighbours in a dictionary. This is because it provides us with a convenient structure to work with, but also because it allows for efficient lookups since dictionaries in Python are hash maps. So while we add each point and its neighbours to the dictionary, we give each point a name, which is x_i where i denotes its place in the set. So each dictionary entry has the following information:

$$\left\{ \text{name: } x_i, \text{ Neighbours: } \{(x_\alpha, y_\alpha), \dots\}, \text{ coordinate: } (x_i, y_i) \right\}$$

To actually construct the discrete Laplacian matrix we do a `list␣comprehension` with parameters exactly as in Remark 2.2.

With the construction of the Vicsek set, the neighbours of each point and the discrete Laplacian, so we move on to generating the eigenfunctions and eigenvalues.

All the constructions mentioned above can be found in `code/VicsekSet/VicsekSet.py` and `code/simulations.py`.

5.1.1 Eigenvalues and eigenfunctions

The way we find eigenvalues is through the `numpy.linalg.eigh` function. The function is for finding eigenvalues and eigenvectors of real symmetric matrices. Some of the best eigenvalue finding algorithms have a runtime of $O(n^2)$, where n is the number of rows in the matrix. As $5^m \cdot 4 + 1$ is a number that grows fast as m increases $(5^m \cdot 4 + 1)^2$ grows even faster making the algorithm slow as m increases.

The code we are describing can be found in `simulations.py` line 95-101 and line 114-133. Now the function `numpy.linalg.eigh` gives us a vector of eigenvalues v_{eig} and a matrix where each column is an eigenvector M_{eig}, having sizes $5^m \cdot 4 + 1 \times 1$ and $5^m \cdot 4 + 1 \times 5^m \cdot 4 + 1$ respectively. The matrix of eigenvectors is actually also our eigenfunctions, since

$$-\Delta_m (M_{\text{eig}})_{i,-} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,-}, \quad \text{so} -\Delta_m (M_{\text{eig}})_{i,j} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,j} = \phi_i(x_j).$$

Thus we have both our eigenvalues and functions.

Calculating the discrete field in each point is the exact same as in Example 3.3, where the white noise W_i is generated by `numpy.random.standard_normal`. To generate the heat maps we plot our points with `matplotlib` and lay in a colorbar where the colour value corresponds to the values from $X_s^m(x)$ for $x \in V_m$.

5.2 The initial goal of the simulations

The point of the simulations was of course to show the convergence of X_s^m to $\widetilde{X_s}$ in distribution, but as we have not been able to simulate much over $m \geq 7$ we don't really see any convergence in distribution. What we can observe instead is that the variance in the fields decreases as s increases.

Figure 5.1 shows 4 different Vicsek sets on V_3 each simulated with their own white noise. As can be seen from the values on the colorbar, the fields decrease in variance as s increases. While this is not difficult to see from the definition of X_s^m, what is interesting is the variance when s is small. If we look at $s = 0.001$, we find that there are many more red spots, meaning there are more transitions between values.

The figure with $s = 50$ it is very much just green and blue. So they have their extreme values, but looking at the colorbar we discover that the reason that they seem so extreme is because the difference in their values is about 10^{161}. Hence the field has almost converged to a single value. This again makes sense looking at X_s^m where the value $(\lambda_j^m)^{-s}$ dominates each term.

5.2.1 Simulations on the Sierpinski gasket

The described theory in this thesis, is as mentioned before, an extension of the paper [4] by Fabrice Baudoin and Li Chen. Therefore the theory that made the simulations of the Vicsek set possible also hold for the Sierpinski gasket. Again, we are not able to simulate m very large, but since the amount of points on the Sierpinski gasket is "only" $\frac{3(3^m+1)}{2}$, we are able to simulate $m \leq 11$. The code for simulations of the Sierpinski gasket are almost identical to the code for the Vicsek set. The main difference being the way the fractal is constructed and the way neighbours are found.

Again we see that the fields, variance converges as s increases, which as explained in the simulations of the Vicsek set, was to be expected.

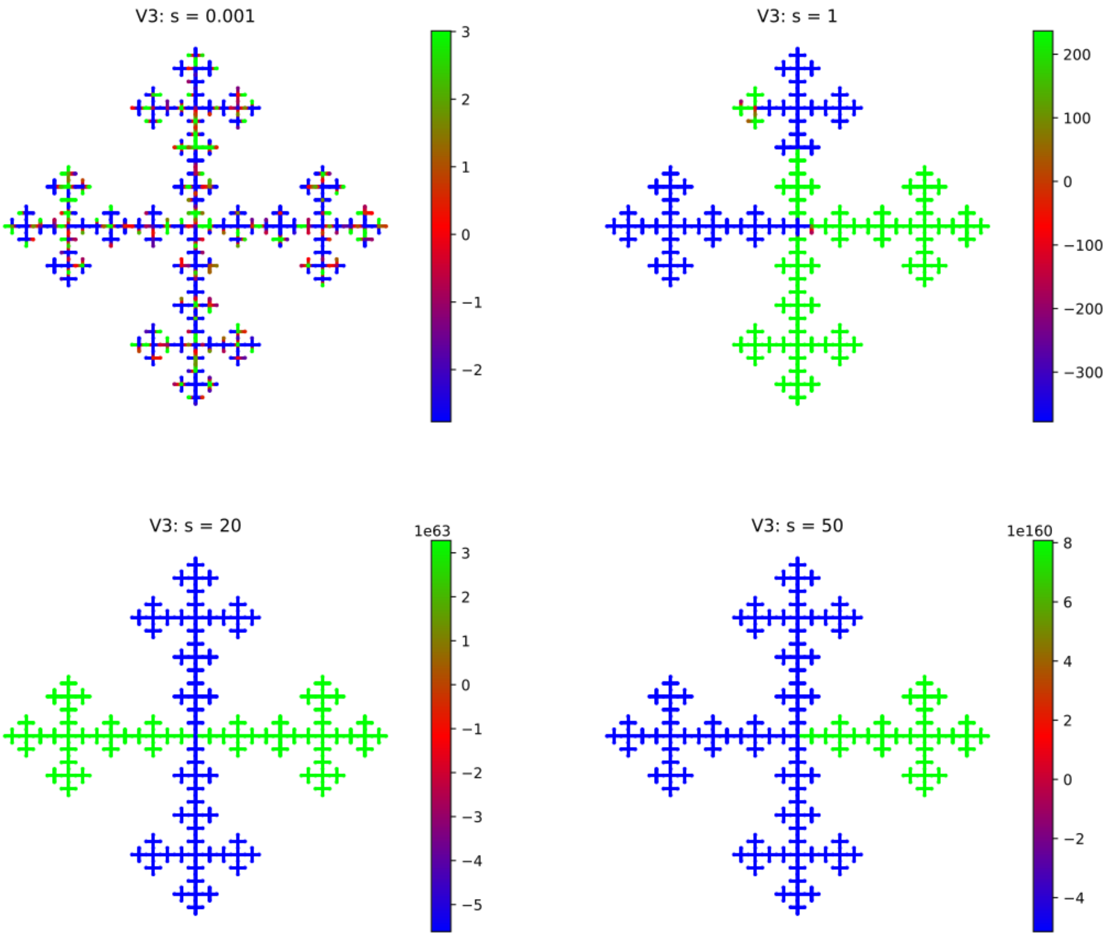

Figure 5.1: Simulations of the Vicsek set on V_3 with different s parameters.

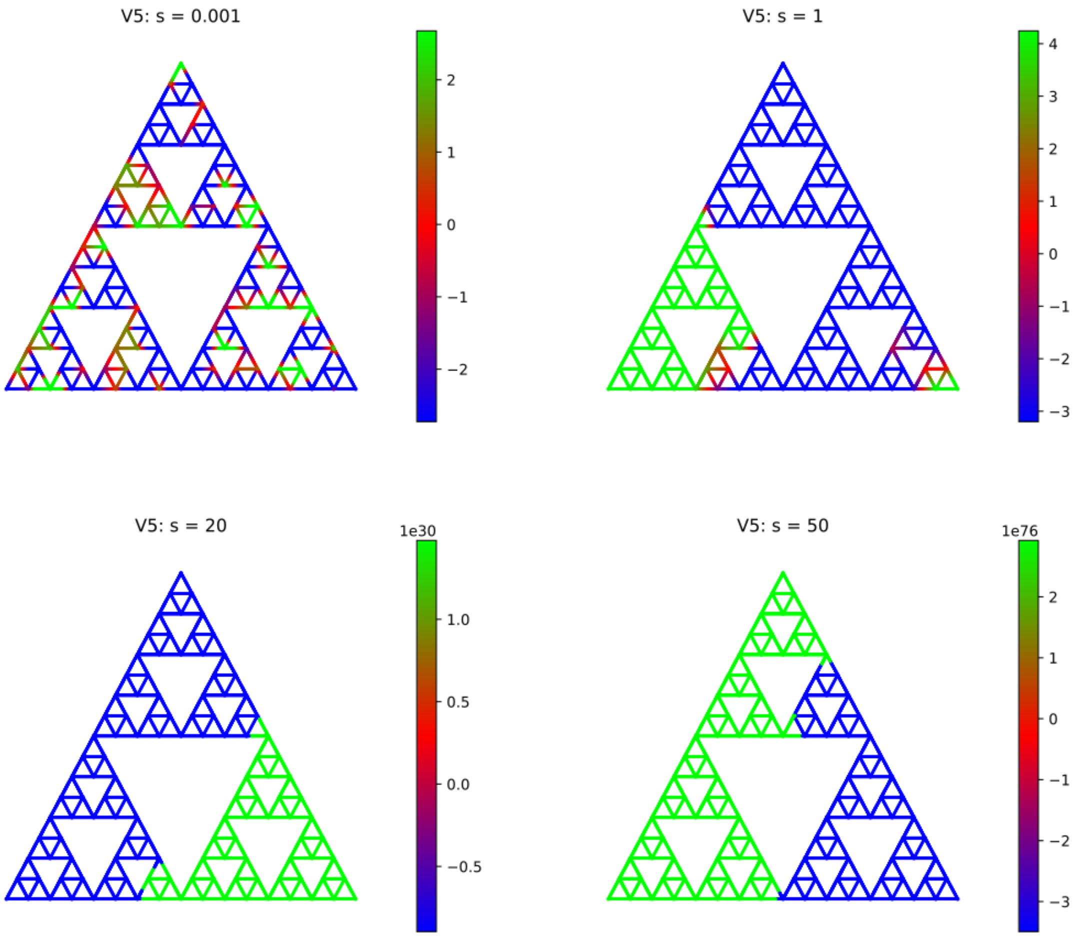

Figure 5.2: Simulations of the Sierpinski set on V_3 with different s parameters.

Bibliography

[1] D. Bakry, *Functional inequalities for markov semigroup*, 2004.

[2] M. T. Barlow and E. A. Perkins, *Brownian motion on the Sierpinski gasket.* Springer-Verlag, 1988, vol. 79.

[3] F. Baudoin, *A short introduction to dirichlet spaces*, 2025.

[4] F. Baudoin and L. Chen, "Dirichlet fractional gaussian fields on the sierpinski gasket and their discrete graph approximations," 2022.

[5] F. Baudoin and L. Chen, "Sobolev spaces and poincaré inequalities on the vicsek fractal," 2022.

[6] F. Baudoin and L. Chen, "Heat kernel gradient estimates for the vicsek set," 2024.

[7] F. Baudoin and C. Lacaux, "Fractional gaussian fields on the sierpinski gasket and related fractals," 2020.

[8] F. Baudoin and C. Lacaux, "Fractional stable random fields on the sierpiński gasket," 2024.

[9] H. Biermé, O. Durieu, and Y. Wang, *Generalized random fields and lévy's continuity theorem on the space of tempered distributions*, 2017.

[10] F. R. K. Chung, *Spectral graph theory*, 2006.

[11] J. Dodziuk, "Eigenvalues of the laplacian and the heat equation," *The American Mathematical Monthly*, vol. 88, no. 9, pp. 686–695, 1981.

[12] K. J. Falconer, *The Geometry of Fractal Sets* (Cambridge Tracts in Mathematics). Cambridge University Press, 1985.

[13] K. Falconer, *Fractal Geometry - Mathematical Foundations and Applications*, English, 2nd. John Wiley and Sons, Ltd, 2003, Second Edition (major revision plus solutions manual).

[14] M. Fukushima and T. Shima, "On a spectral analysis for the sierpinski gasket," 1992.

[15] R. A. van de Geijn, *Notes on the symmetric qr algorithm*, 2014.

[16] A. Gervani, "Hausdorff dimension and iterated function systems," Bachelor Thesis, Dipartimento di Matematica "Tullio Levi-Civita", Universit'a degli Studi di Padova, Sep. 2023.

[17] M. Hazeem, "Similarities and hausdor dimension in fractal geometry," Matematiska Institutionen, Stockholms Universitet, Självständiga arbeten i matematik, Jan. 2022.

[18] J. T. Henrikson, *Completeness and total boundedness of the hausdorff metric*, 1999.

[19] T. R. Johansen, "The bochner – minlos theorem for nuclear spaces and an abstract white noise space," 2003.

[20] J. Kigami, *Analysis on Fractals* (Cambridge Tracts in Mathematics). Cambridge University Press, 2001.

[21] T. G. Kurtz, “Extensions of trotter’s operator semigroup approximation theorems,” *Journal of Functional Analysis*, vol. 3, no. 3, pp. 354–375, 1969.

[22] J.-F. Le Gall, *Brownian Motion, Martingales, and Stochastic Calculus*. Jan. 2016, vol. 274.

[23] D. N. Martin T. Barlow, *Lectures on Probability Theory and Statistics*. Springer Berlin, Heidelberg, 1998, pp. 1–121, Ecole d’Ete de Probabilites de Saint-Flour XXV - 1995.

[24] A. Pazy, *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer New York, NY, 1992.

[25] A. Pietsch, *Über die Erzeugung von (F)-Räumen durch selbstadjungierte Operatoren*. German. Springer, 1966.

[26] E. Skibsted, *Notes for a course in advancd analysis*. Institut for Matematik, Aarhus Universitet, 2023.

[27] R. S. Strichartz, *Differential Equations on Fractals: A Tutorial*. Princeton University Press, 2006.

[28] M. Taylor, *The spectral theorem for self-adjoint and unitary operators*.

[29] S. Thorbjørnsen, *Forelæsningsnoter i Videregående Sandsynlighedsteori*. Institut for Matematik, Aarhus Universitet, 2022.

[30] S. Thorbjørnsen, *Introduction to Measure and Integration Theory*. Institut for Matematik, Aarhus Universitet, 2024.

[31] F. Trèves, *Topological Vector Spaces, Distributions and Kernels*. Academic Press, 1967.

[32] J. B. Walsh, “An introduction to stochastic partial differential equations,” in *École d’Été de Probabilités de Saint Flour XIV - 1984*, P. L. Hennequin, Ed., Berlin, Heidelberg: Springer Berlin Heidelberg, 1986, pp. 265–439.